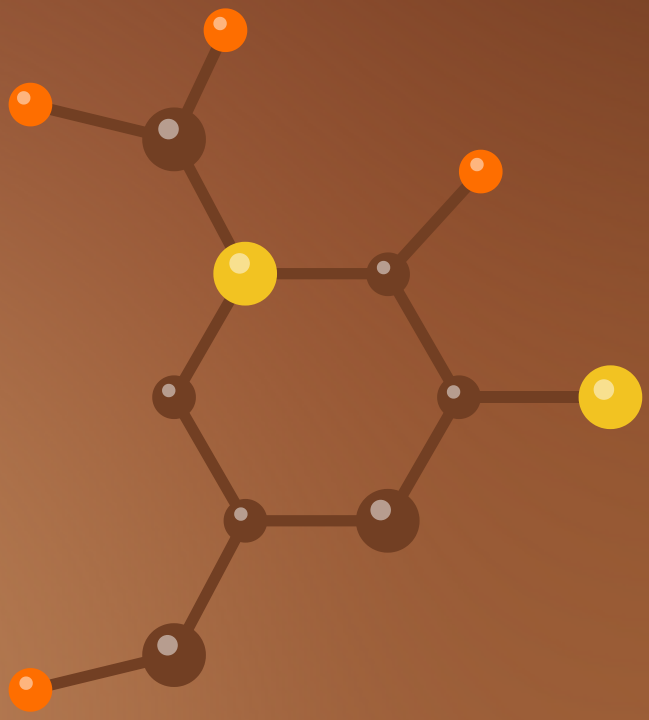
ALLEN CAREER INSTITUTE
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ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

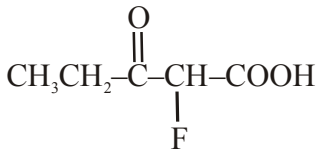
Carboxylic Acids & its Derivative, Alphatic Amines

English Medium

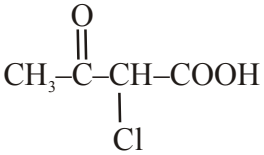
CARBOXYLIC ACIDS AND IT'S DERIVATIVE, ALIPHATIC AMINES

EXERCISE # O-I

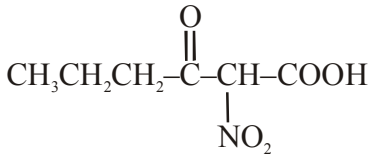
1. Correct order of decarboxylation



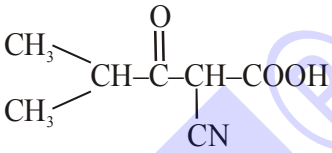
(a)



(b)



(c)



(d)

(A) a > b > c > d (B) c > d > b > a (C) c > d > a > b (D) d > c > a > b

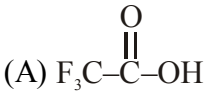
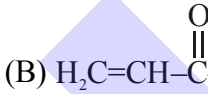
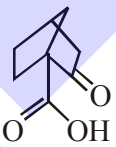
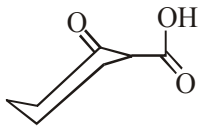
CA0013

2. Which will form lactone on treatment with NaOH ?

(A) α-Bromo acid (B) β-Bromo acid (C) β-Hydroxy acid (D) δ-Bromo acid

CA0011

3. Which of the following carboxylic acids undergo decarboxylation simply on heating ?

(A)  (B)  (C)  (D) 

CA0039

4. $CH_3COONa \xrightarrow[\Delta]{NaOH, CaO}$ Product (P)

Choose the incorrect option?

(A) Reaction is known as decarboxylation.

(B) Carbanion is formed as an intermediate.

(C) When aq. solution of reactant is electrolysed methane will be obtained at anode

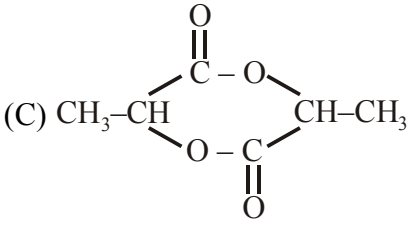
(D) P is methane gas

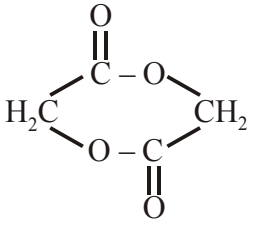
CA0151

5. 

P is :

(A) CH_3COOH (B) CH_3OH

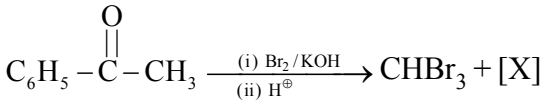
(C) 

(D) 

CA0152

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6. In the given reaction,

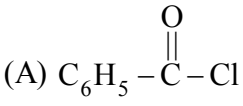


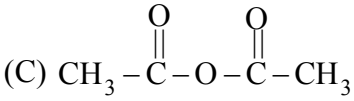
[X] will be :

(A) C_6H_5–CHO (B) C_6H_5COOH (C) C_6H_5–CH_2OH (D) CH_3COOH

CA0003

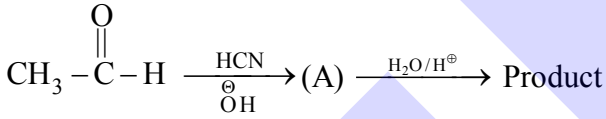
7. Which one of the following compounds gives carboxylic acid with HNO_2?

(A)  (B) $C_6H_5CONH_2$

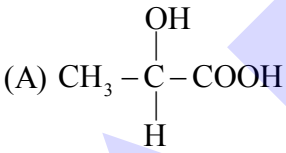
(C)  (D) $CH_3COOC_2H_5$

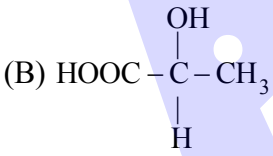
CA0007

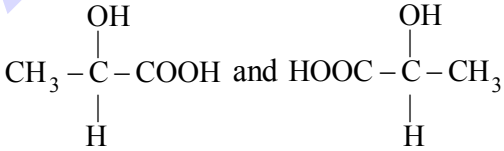
8. In the reaction sequence ,

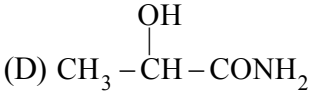


Product will be :

(A) 

(B) 

(C) Mixture of 

(D) 

CA0008

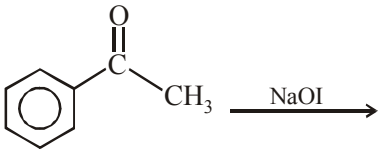
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

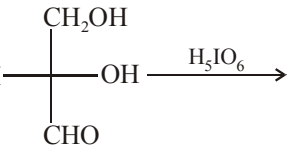
9. Which of the following compounds will not give acetic acid with $KMnO_4/H^{\oplus}/\Delta$:

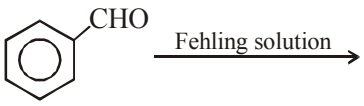
(A) $CH_3–CHO$ (B) $CH_3–CH=CH–CH_3$

(C) $CH_3–C\equiv C–CH_3$ (D) CH_3OH **CA0153**

10. In which of the following reaction carboxylic acid/salt is not obtained as one of the major product

(A) 

(B) 

(C) 

(D) 

CA0154

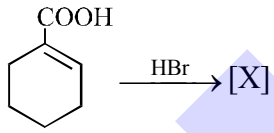
11. Which optically active compound on reduction with $LiAlH_4$ will give optically inactive compound?

(A) $CH_3-\underset{\displaystyle OCH_3}{\underset{|}{CH}}-COOH$ (B) $CH_3-CH_2-\underset{\displaystyle OH}{\underset{|}{CH}}-COOH$

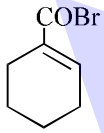
(C) $CH_3-CH_2-\underset{\displaystyle CH_2OH}{\underset{|}{CH}}-COOH$ (D) $CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2-COOH$

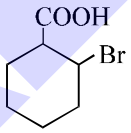
CA0010

12. In the given reaction:

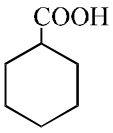


[X] will be :

(A) 

(B) 

(C) 

(D) 

CA0012

13. Arrange these esters in decreasing order of ease of esterfication with $CH_3OH/H^{\oplus}$:

(I) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$ (II) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-COOH$

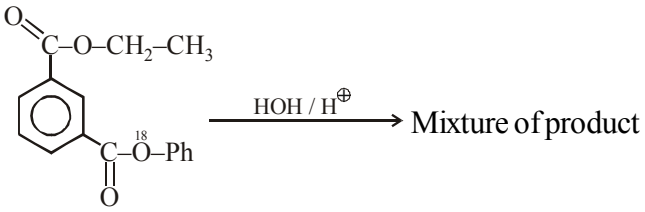
(III) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-COOH$ (IV) $(CH_3–CH_2)_3C–COOH$

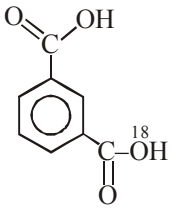
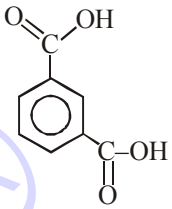
(A) II > I > III > IV (B) I > II > III > IV (C) III >IV > II > I (D) IV > III > II > I

CA0009

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14. Which of the following will not be present in mixture of product :



(A) $CH_3–CH_2–OH$ (B)  (C) $Ph–\overset{18}{O}H$ (D) 

CA0155

15. RCOOR' cannot be prepared by:

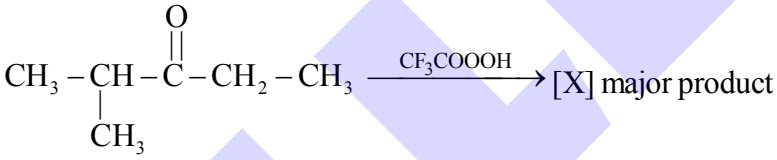
(A) Esterification of RCOOH

(B) Reaction of $CH_3CH=CH_2$ with methanol

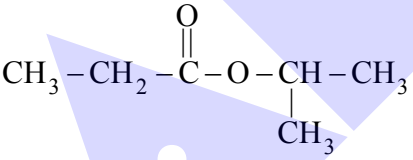
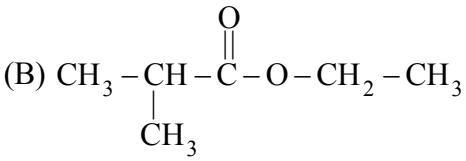
(C) Baeyer-Villiger oxidation of RCOR' with peroxy acid

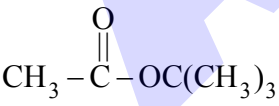
(D) reaction of RCOCl with R'OH **CA0156**

16. In the given reaction:

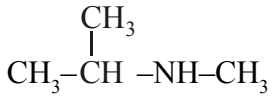
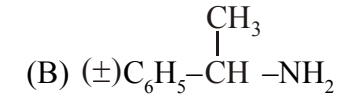


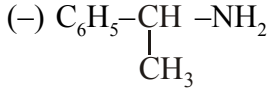
[X] will be:

(A)  (B) 

(C)  (D) $(CH_3)_3COOCH_3$ **CA0157**

17. Which of the following amines could be used as a resolving agent for a mixture of carboxylic acid into respective enantiomer ?

(A)  (B) 

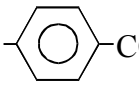
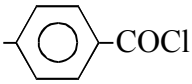
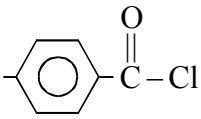
(C)  (D) CH_3NH_2

CA0158

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18. Arrange following compounds in decreasing order of reactivity for hydrolysis reaction :

(I) C_6H_5COCl (II)  (III)  (IV) 

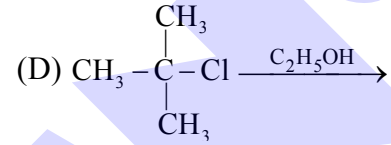
(A) II > IV > I > III (B) II > IV > III > I

(C) I > II > III > IV (D) IV > III > II > I

CA0006

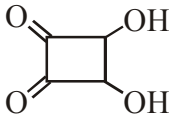
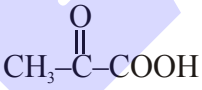
19. In which reaction product is hydrocarbon ?

(A) $RCOOK \xrightarrow{\text{Electrolysis}}$ (B) $RCOOAg \xrightarrow{I_2/\Delta}$

(C) $CH_3–CH_3 \xrightarrow{Cl_2/h\nu}$ (D) 

CA0001

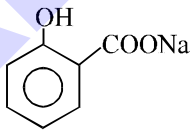
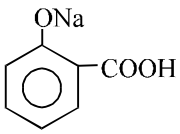
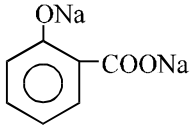
20. Which of the following can released CO_2 with $NaHCO_3$.

 (i) $CH_2(COOH)_2$ (ii)  (iii)

(A) (i), (ii) & (iii) (B) (i) & (ii) (C) (ii) & (iii) (D) (i) & (iii)

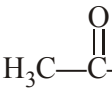
CA0018

21. Sodium bicarbonate reacts with salicylic acid to form :

(A) C_6H_5ONa (B)  (C)  (D) 

CA0019

22. Among the following, which statement is not correct ?

(A)  will not respond to haloform test

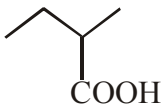
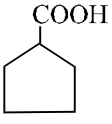
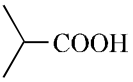
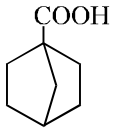
(B) Schiff's regent and Schiff's base are different compounds

(C) Fehling's solution is a good reagent to detect aromatic aldehydes

(D) Both aldehyde and ketone can react with 2,4-dinitrophenylhydrazise reagent **CA0159**

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23. Which one of the following compounds will not give HVZ reaction?

(A)  (B)  (C)  (D) 

CA0160

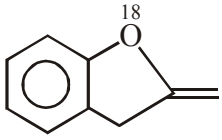
24. In the given reaction :

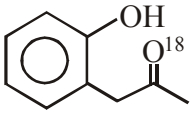
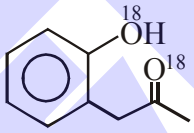
$$CH_3CHO \xrightarrow[(ii)\ H_2O/H^{\oplus}/\Delta]{(i)\ NaCN/HCl} (A) \xrightarrow[Reagent]{Fenton's} (B)$$

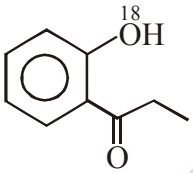
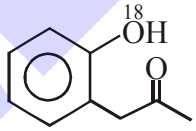
(B) will be :

(A) Acetic acid (B) Oxalic acid (C) Pyruvic acid (D) Citric acid

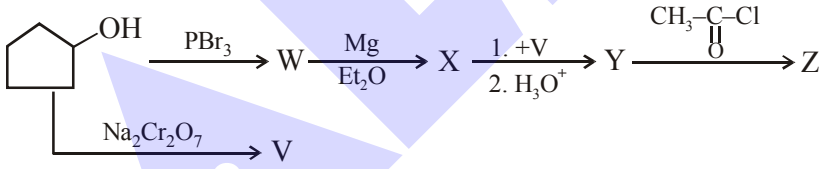
CA0025

25.  ; Major organic product

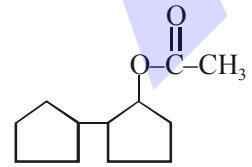
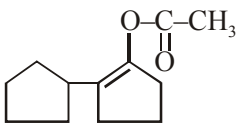
(A)  (B) 

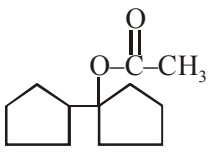
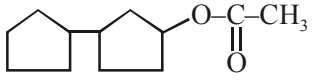
(C)  (D) 

CA0161

26. 

Product Z of the above reaction is :

(A)  (B) 

(C)  (D) 

CA0162

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27. In the given reaction :

[X] + Acetic anhydride $\longrightarrow$ Aspirin

[X] will be :

(A) Benzoic acid

(B) *o*-methoxybenzoic acid

(C) *o*-Hydroxybenzoic acid

(D) *p*-Hydroxybenzoic acid

CA0005

28. Which of the following will not form acetyl chloride with PCl_5?

(A) MeCOOH

(B) MeCOOMe

(C) MeCOOCOMe

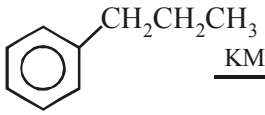
(D) Me–$CONH_2$

CA0163

29. Which of the following set of reaction can not prepare carboxylic acid as the final product :

(A) $R–MgX + O=C=O \xrightarrow{\text{dry ether}} A \xrightarrow{H_3O^+}$

(B) $R–CN \xrightarrow[\text{(ii) } H_2O / H^+]{\text{(i) } SnCl_2 + HCl}$

(C) 

(D) $CH_3(CH_2)_8CH_2OH \xrightarrow[H_2SO_4]{CrO_3}$

CA0002

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EXERCISE # O-II

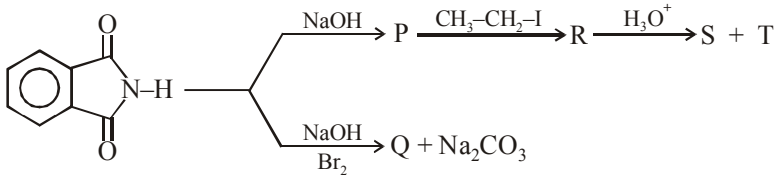
1. $RCH_2CONH_2 + NaOBr \longrightarrow ?$

Rate of reaction will be faster if 'R' is

(A) CH_3- (B) C_2H_5- (C) NO_2- (D) $CN-$

CA0038

2.



If T can evolve effervescence of CO_2 with a $NaHCO_3$, then incorrect statement is :

(A) S & Q can be distinguished by dye azo test

(B) T is most acidic among all isomeric benzenoid dicarboxylic acid

(C) Q & S can be distinguished by mustered oil test

(D) P, Q & T all are soluble in a $NaHCO_3$ **CA0164**

3. N-Ethyl pthalimide on hydrolysis gives:

(A) Methyl alcohol (B) Ethyl amine

(C) Dimethyl amine (D) Diethyl amine

CA0014

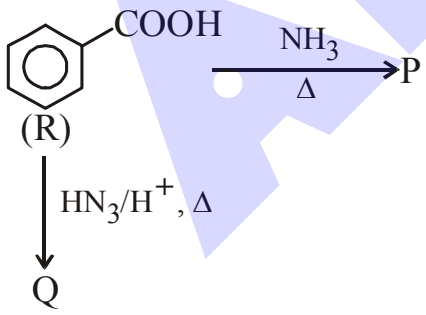
4. $C_6H_5CONH_2 \xrightarrow{Br_2/OD^{\ominus}/D_2O} P$, 'P' is :

(A) $C_6H_5COND_2$ (B) $C_6H_5ND_2$

(C) C_6H_5NHD (D) $C_6H_5NH_2$

CA0022

5.



The correct statement is/are :

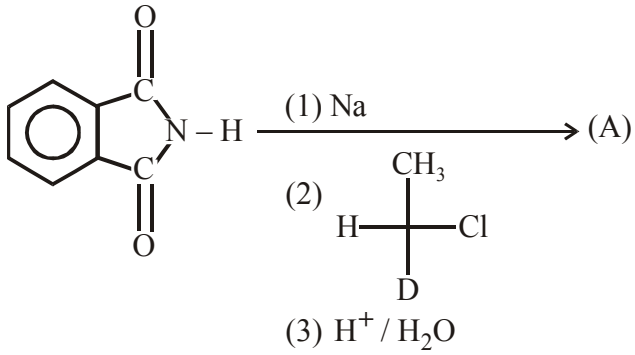
(A) Formation of Q is schmmidt reaction

(B) Electron density in ring : (R) > (Q) > (P)

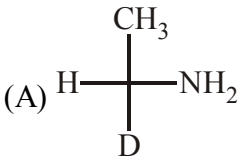
(C) Q is less basic than P

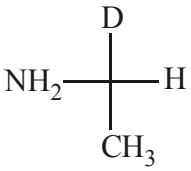
(D) Formation of P is Hoffmann bromamide degradation. **CA0165**

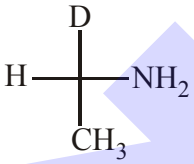
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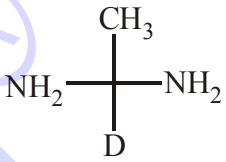
6. 

Product A is :

(A) 

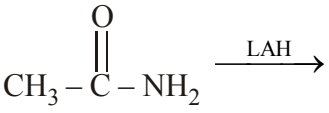
(B) 

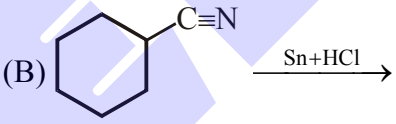
(C) 

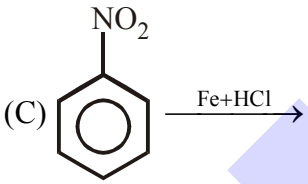
(D) 

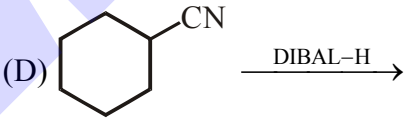
CA0166

7. Which of the following reaction will not produce 1° amine?

(A) 

(B) 

(C) 

(D) 

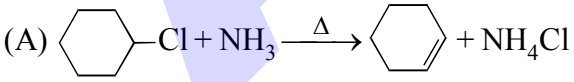
CA0167

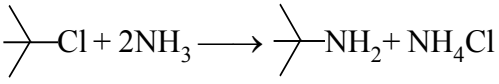
8. Hofmann degradation is given by:

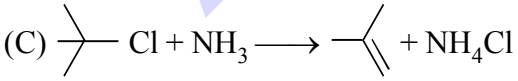
(A) Acetophenone (B) Acid chloride (C) Acid anhydride (D) Acetamide

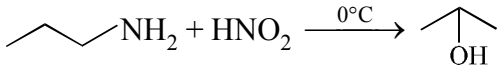
CA0041

9. Which is incorrect reaction:

(A) 

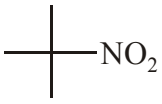
(B) 

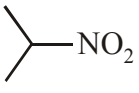
(C) 

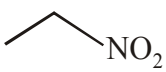
(D) 

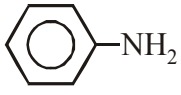
CA0168

10. Which of the following compound will not react with HNO_2 :

(A) 

(B) 

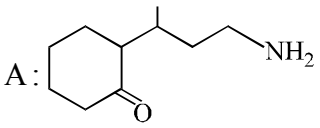
(C) 

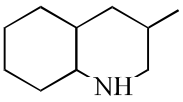
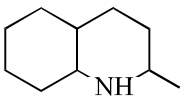
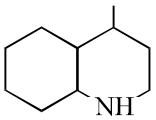
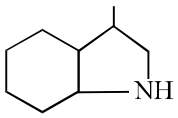
(D) 

CA0169

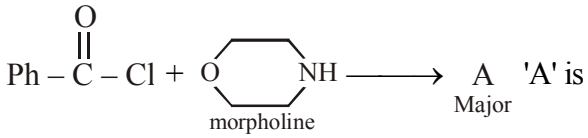
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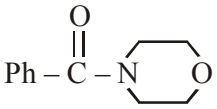
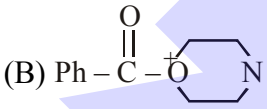
11. Reductive amination of A forms:

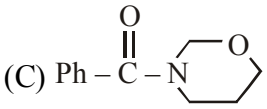
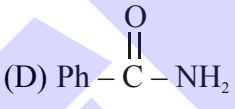
A : 

(A)  (B)  (C)  (D) 

CA0004

12. 

(A)  (B) 

(C)  (D) 

CA0016

13. $CH_3NH_2 \xrightarrow{(CH_2)_2O} (A)$

(A) 'A' is more basic than CH_3NH_2 (B) 'A' is less basic than CH_3NH_2

(C) 'A' is Ter-amine (D) None

CA0017

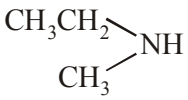
14. Which of the following diazonium salt is relatively stable at 0-5°C:

(A) $CH_3–N\equiv N\}^{\oplus}Cl^-$ (B) $(CH_3)_2CH–N\equiv N\}^{\oplus}Cl^-$

(C) $C_6H_5–N\equiv N\}^{\oplus}Cl^-$ (D) $(CH_3)_3C–N\equiv N\}^{\oplus}Cl^-$

CA0020

15. Which is most volatile ?

(A) $CH_3CH_2CH_2NH_2$ (B) $(CH_3)_3N$ (C)  (D) CH_3OH

CA0021

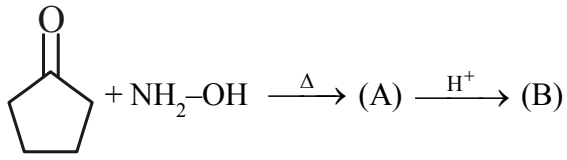
16. In the chemical reaction, $CH_3CH_2NH_2 + CHCl_3 + 3KOH \rightarrow (A) + (B) + 3H_2O$, the compounds (A) and (B) are respectively - **[AIEEE-2007]**

(A) C_2H_5CN and 3KCl (B) $CH_3CH_2\ CONH_2$ and 3KCl

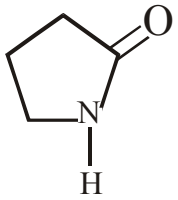
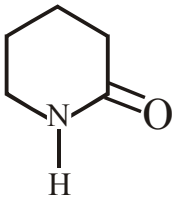
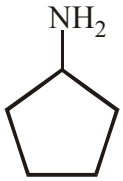
(C) C_2H_5NC and K_2CO_3 (D) C_2H_5NC and 3KCl

CA0069

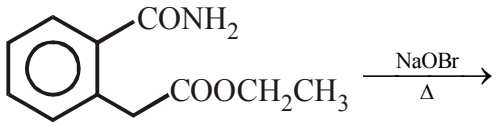
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Bimole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbon\01_Th & Ex.p65

17. 

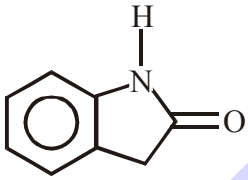
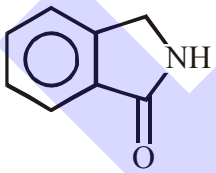
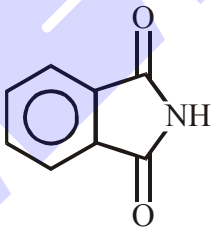
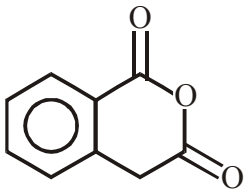
B is?

(A)  (B)  (C)  (D) None of the above

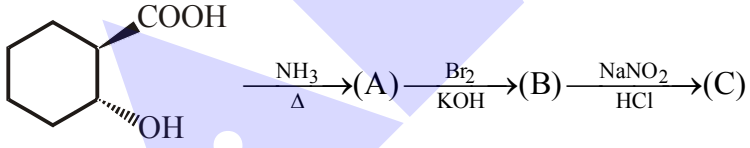
CA0170

18. 

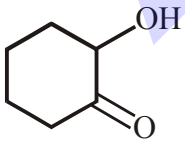
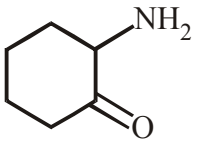
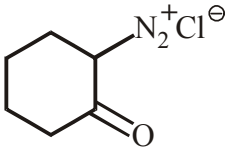
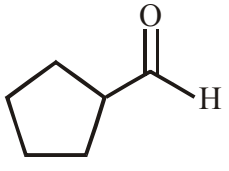
The final product of the reaction is :

(A)  (B)  (C)  (D) 

CA0171

19. 

Final product C is :-

(A)  (B)  (C)  (D) 

CA0172

20. Which among the following has highest boiling point?

(A) $n\text{–}C_4H_9NH_2$ (B) $(C_2H_5)_2NH$ (C) $C_2H_5N(CH_3)_2$ (D) $C_2H_5CH(CH_3)_2$

CA0173

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21. Correct order of basic strength in aq. medium is :-

(A) $(C_2H_5)_2NH > C_2H_5NH_2 > (C_2H_5)_3N$ (B) $(CH_3)_2NH > (CH_3)_3N > CH_3NH_2$

(C) $(C_2H_5)_3N > (C_2H_5)_2NH > C_2H_5NH_2$ (D) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

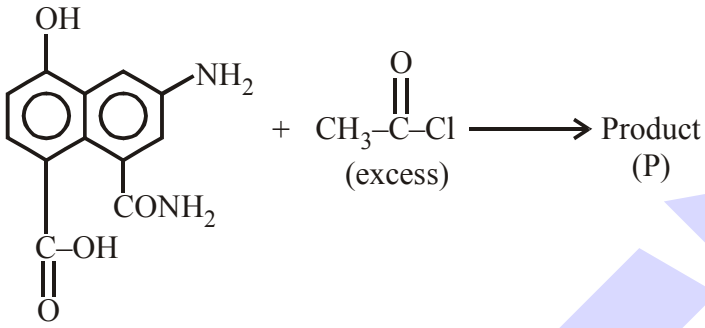
CA0174

22. Correct order of solubility is :

(A) $C_6H_5NH_2 > (C_2H_5)_2NH > C_2H_5NH_2$ (B) $(C_2H_5)_2NH > C_6H_5NH_2 > C_2H_5NH_2$

(C) $C_2H_5NH_2 > (C_2H_5)_2NH > C_6H_5NH_2$ (D) None of the above **CA0175**

23.



(A)

Increass in molecular mass from A to P is?

(A) 86 (B) 168 (C) 84 (D) 172

CA0176

24. $CH_3–\underset{\underset{NH_2}{|}}{CH}–CO_2H$ can be converted into $CH_3–\underset{\underset{NH_2}{|}}{CH}–CH_2OH$ by the use of :

(A) H_2(Pd) (B) $LiAlH_4$ (C) $NaBH_4$ (D) CH_3MgBr

CA0177

25. RNH_2 and R_2NH can be distinguished by

(A) HCl (B) H_2SO_4 (C) Na (D) $CS_2 + HgCl_2$

CA0178

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EXERCISE # S-I

1. Mixture of 1°, 2° and 3° amines can be separated by:

(A) Hinsberg's method
(B) Hofmann's isocyanide test
(C) Fractional distillation
(D) $NaNO_2$ HCl

CA0027

2. Which of the following amine reacts with Hinsberg reagent to give base soluble product :-

(A) Neopentyl amine
(B) sec propyl amine
(C) diethyl amine
(D) ethyl methyl amine

CA0029

3. Acetic anhydride and ammonia gives the product:

(A) CH_3CONH_2
(B) $CH_3CONHCH_3$
(C) CH_3CN
(D) CH_3COONH_4

CA0035

4. $CH_3-\overset{\overset{O}{\|}}{C}-O-CH_2-CH_3 \underset{}{\overset{^{18}HOH / H^{\oplus}}{\rightleftharpoons}}$ Ethanoic acid + Ethanol

Isotopic oxygen of water will be present with

(A) Ethanoic acid
(B) Ethanol
(C) After some time it will also be present in some molecules of ester
(D) None of these

CA0036

5. The presence of alphatic primary amine can be confirmed by its reaction with :

(A) HNO_2 followed by CAN solution
(B) $CHCl_3$ + NaOH
(C) CS_2 & $HgCl_2$
(D) H_2SO_4

CA0042

6. $H_3C-\overset{\overset{O}{\|}}{C}-O-\underset{\underset{Ph}{|}}{C}=CH_2 + H_2O \xrightarrow{H_3O^+} X + Y$

X and Y are :

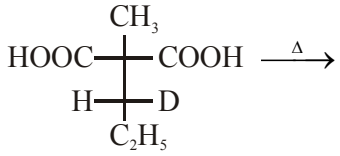
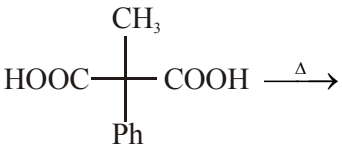
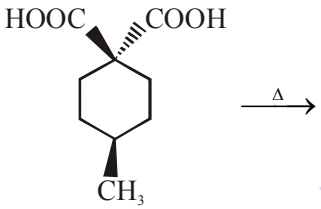
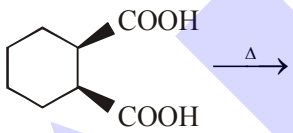
(A) X is $CH_3-\overset{\overset{O}{\|}}{C}-O-CH_2-Ph$
(B) Y is PhCHO
(C) Y is $H_3C-\overset{\overset{O}{\|}}{C}-OH$
(D) X is $Ph-\overset{\overset{O}{\|}}{C}-CH_3$

CA0045

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Matching Type Questions

7. Match the following question :

Column - I (Reaction)	**Column-II (Products)**
(A) 	(P) Diastereomers
(B) 	(Q) Racemic mixture
(C) 	(R) Meso comp.
(D) 	(S) CO_2 gas will evolve

CA0047

8. Match the following question :

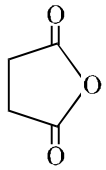
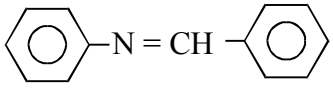
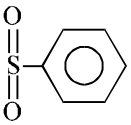
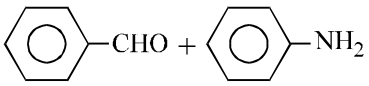
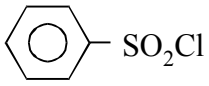
Column I (Organic compounds oxidised by HIO_4)	**Column II (Products of HIO_4 oxidation)**
(A) CH_3COCHO	(P) PhCH = O + HCOOH
(B) 1,2-cyclohexane dione	(Q) $CH_3CH_2CHO + HOOCCH_3$
(C) PhCH (OH) CHO	(R) $HOOC\ (CH_2)_4COOH$
(D) $CH_3CH_2CH\ (OH)\ COCH_3$	(S) $CH_3COOH + HCOOH$

CA0048

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9. Match the following question :

Column I (Reactions)	**Column II (Products)**
(A) CH_3NH_2 + 	(P) 
(B) $(CH_3)_2NH + C_6H_5COCl$	(Q) $CH_3 - NH-$ 
(C) 	(R) $(CH_3)_2N-C$ 
(D) CH_3NH_2 + 	(S) $CH_3NH-C-CH_2CH_2-C-O^-$

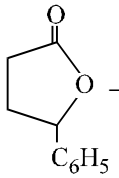
CA0049

10. Match the following question :

Column I (Correct about product)	**Column II**
(A) $R - CN \xrightarrow{H_3O^+}$	(P) Product is yellow oily liquid
(B) $R - NH_2 \xrightarrow[KOH]{CHCl_3}$	(Q) Gives red colour with CAN
(C) $R - NH_2 \xrightarrow[HCl]{NaNO_2}$	(R) Gives fruity smell with CH_3OH
(D) $R_2NH \xrightarrow[HCl]{NaNO_2}$	(S) Foul smelling compound is formed.

CA0050

11. Match the following question :

Column I	**Column II**
(A) $PhCONH_2 \longrightarrow Ph-C\equiv N$	(P) P_2O_5
(B)  $\longrightarrow CH_2OH-(CH_2)_2-CHOH-C_6H_5$	(Q) $LiAlH_4$
(C) $CH_3-\underset{\underset{O}{\|}}{C}-OCH_3 \longrightarrow CH_3CH=O$	(R) $H_2/Pd-BaSO_4$
(D) $CH_3COCl \longrightarrow CH_3-CHO$	(S) DIBALH

CA0051

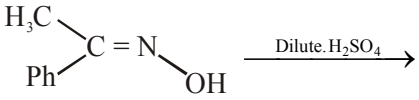
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12. Match the following question :

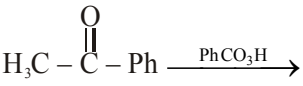
Column I	Column II
(A) $RCN \xrightarrow{\text{reduction}}$	(P) 1° Amine
(B) $RCN \xrightarrow[(ii)\,H_2O]{(i)\,CH_3MgBr}$	(Q) Alcohol
(C) $RNC \xrightarrow{\text{hydrolysis}}$	(R) Ketone
(D) $RNH_2 \xrightarrow{HNO_2}$	(S) Acid

CA0052

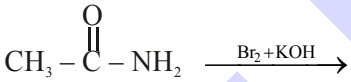
13. Find out number of reactions which involve electron dificient nitrogen [Nitrene character] during reaction mechanism.

(a) 

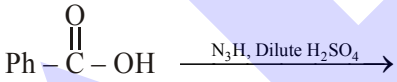
CA0053

(b) 

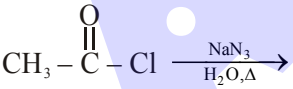
CA0054

(c) 

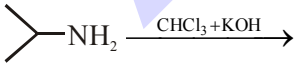
CA0055

(d) 

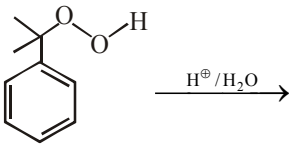
CA0056

(e) 

CA0057

(f) 

CA0058

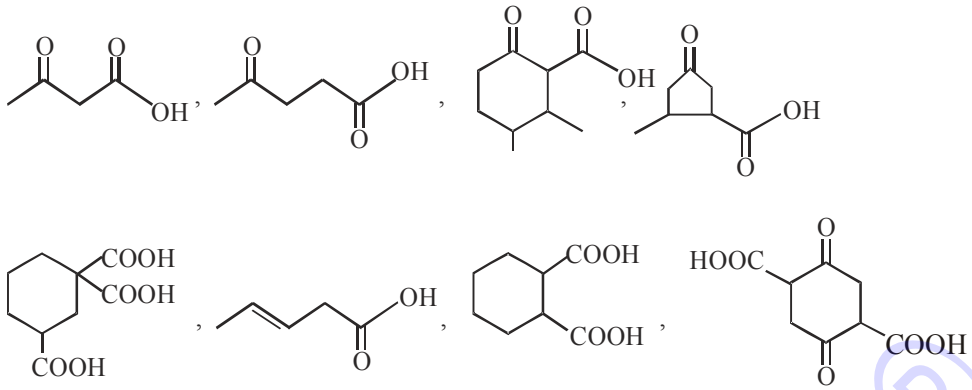
(g) 

CA0059

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14. Examine the structure of following compounds, and find out number of compounds that will undergo decarboxylation in presence of heat.



CA0061

15. $CH_3CH_2-\underset{\underset{NH_2}{|}}{CH}-COOH \xrightarrow{\Delta}$ P (no. of products)

$CH_3CH_2-\underset{\underset{OH}{|}}{CH}-COOH \xrightarrow{\Delta}$ Q (no. of products)

$CH_3\underset{\underset{OH}{|}}{CH}-CH_2-COOH \xrightarrow{\Delta}$ R (no. of products)

$\underset{\underset{OH}{|}}{CH_2}-CH_2-CH_2-COOH \xrightarrow{\Delta}$ S (no. of products)

In all reactions the sum of product is.

CA0062

How will you bring about the following transformation:

16. Propanoic acid into lactic acid.

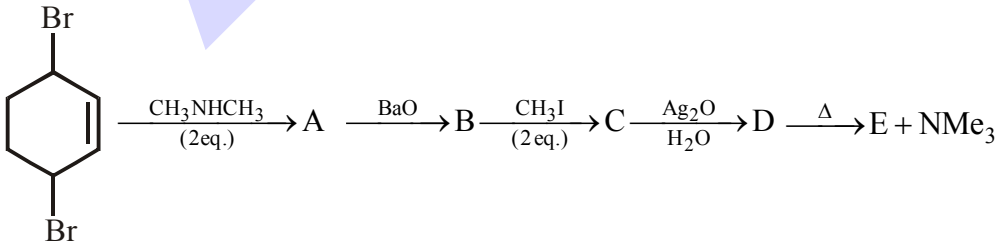
CA0063

17. Ethyl benzene to 2-phenyl propionic acid.

CA0064

18. Acetamide from acetone.

CA0065

19. 

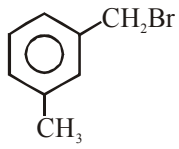
Number of sp^2 hybridised orbital present in E is?

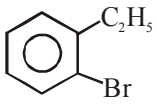
CA0179

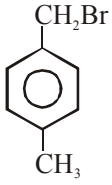
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

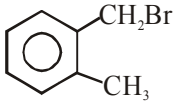
EXERCISE # JEE-MAIN

1. Compound (A), C_8H_9Br, gives a white precipitate when warmed with alcoholic $AgNO_3$. Oxidation of (A) gives an acid (B), $C_8H_6O_4$. (B) easily forms anhydride on heating. Identify the compound (A): **[AIEEE-2013]**

(1) 

(2) 

(3) 

(4) 

CA0072

2. An organic compound A upon reacting with NH_3 gives B. On heating, B gives C. C in presence of KOH reacts with Br_2 to give $CH_3CH_2NH_2$. A is :- **[AIEEE-2013]**

(1) CH_3COOH (2) $CH_3CH_2CH_2COOH$ (3) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH$ (4) CH_3CH_2COOH

CA0073

3. In the reaction $CH_3COOH \xrightarrow{LiAlH_4} A \xrightarrow{PCl_5} B \xrightarrow{alc.KOH} C$ 'C' is **[AIEEE-2014]**

(1) Ethylene (2) Acetyl chloride (3) Acetaldehyde (4) Acetylene.

CA0075

4. In the presence of a small amount of phosphorous, aliphatic carboxylic acids react with chlorine or bromine to yield a compound in which α - hydrogen has been replaced by halogen. This reaction is known as : **[JEE(Main)-2015]**

(1) Etard reaction (2) Hell - Volhard - Zelinsky reaction

(3) Wolff - Kischner reaction (4) Rosenmund reaction

CA0076

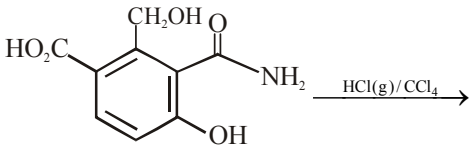
5. In the Hofmann bromamide degradation reaction, the number of moles of NaOH and Br_2 used per mole of amine produced are : **[JEE(Main)-2016]**

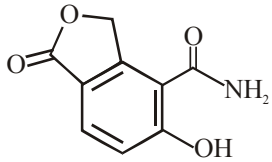
(1) Four moles of NaOH and one mole of Br_2 (2) One mole of NaOH and one mole of Br_2

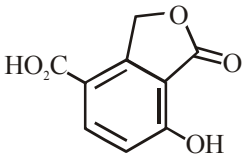
(3) Four moles of NaOH and two moles of Br_2 (4) Two moles of NaOH and two moles of Br_2

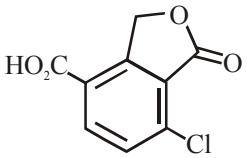
CA0077

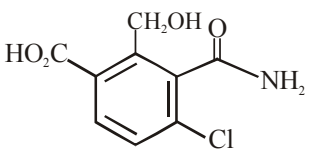
6. The major product expected from the following reaction is : **[JEE(Main On-Line)-2017]**



(1) 

(2) 

(3) 

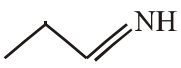
(4) 

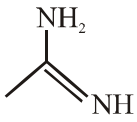
CA0078

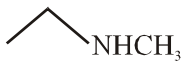
node06\B0BC BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Bimole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbon\01_Th & Ex.p65

7. The increasing order of basicity of the following compounds is : **[JEE(Main)-2018]**

(a) 

(b) 

(c) 

(d) 

(1) (b) < (a) < (c) < (d)

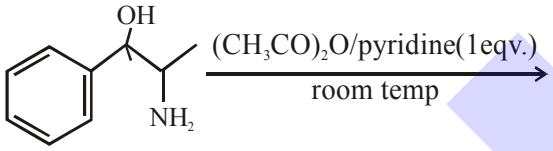
(2) (b) < (a) < (d) < (c)

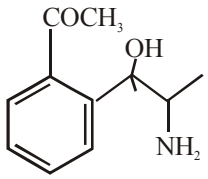
(3) (d) < (b) < (a) < (c)

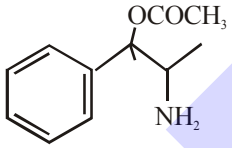
(4) (a) < (b) < (c) < (d)

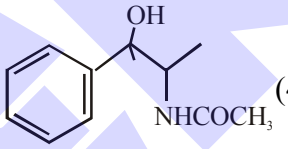
CA0079

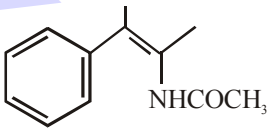
8. The major product obtained in the following reaction is : **[JEE Main (Jan)-2019]**



(1) 

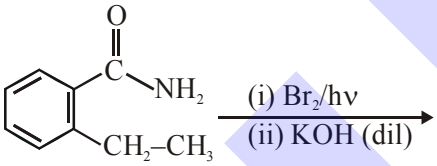
(2) 

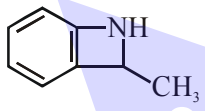
(3) 

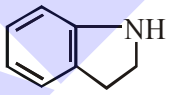
(4) 

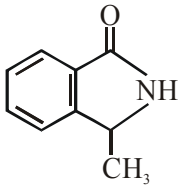
CA0081

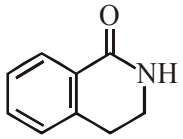
9. The major product of the following reaction is : **[JEE Main (Jan)-2019]**



(1) 

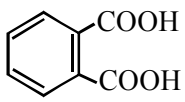
(2) 

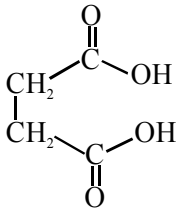
(3) 

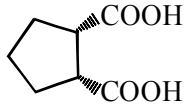
(4) 

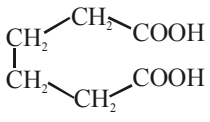
CA0082

10. Which dicarboxylic acid in presence of a dehydrating agent is least reactive to give an anhydride : **[JEE Main (Jan)-2019]**

(1) 

(2) 

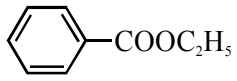
(3) 

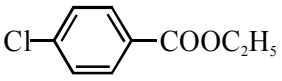
(4) 

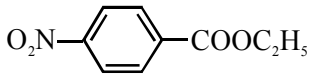
CA0083

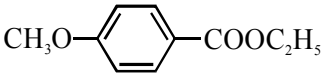
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

11. The decreasing order of ease of alkaline hydrolysis for the following esters is : **[JEE Main (Jan)-2019]**


I


II

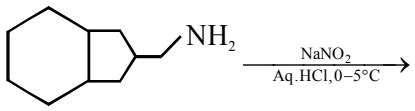

III

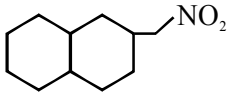
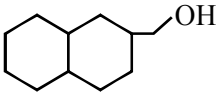
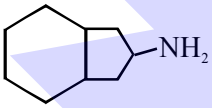
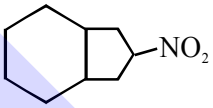

IV

(1) IV > II > III > I (2) III > II > I > IV (3) III > II > IV > I (4) II > III > I > IV

CA0084

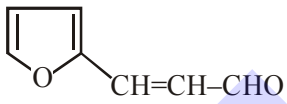
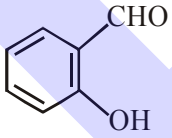
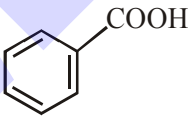
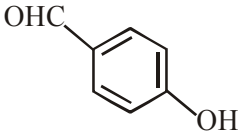
12. The major product formed in the reaction given below will be : **[JEE Main (Jan)-2019]**



(1)  (2)  (3)  (4) 

CA0085

13. An aromatic compound 'A' having molecular formula $C_7H_6O_2$ on treating with aqueous ammonia and heating forms compound 'B'. The compound 'B' on reaction with molecular bromine and potassium hydroxide provides compound 'C' having molecular formula C_6H_7N. The structure of 'A' is : **[JEE Main (Jan)-2019]**

(1)  (2)  (3)  (4) 

CA0087

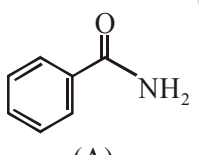
14. A compound 'X' on treatment with Br_2/NaOH, provided C_3H_9N, which gives positive carbylamine test. Compound 'X' is :- **[JEE Main (Jan)-2019]**

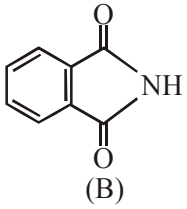
(1) $CH_3COCH_2NHCH_3$ (2) $CH_3CH_2COCH_2NH_2$

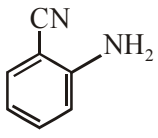
(3) $CH_3CH_2CH_2CONH_2$ (4) $CH_3CON(CH_3)_2$

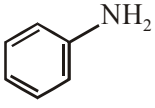
CA0088

15. The increasing order of reactivity of the following compounds towards reaction with alkyl halides directly is : **[JEE Main (Jan)-2019]**


(A)


(B)


(C)


(D)

(1) (B) < (A) < (D) < (C) (2) (B) < (A) < (C) < (D)

(3) (A) < (C) < (D) < (B) (4) (A) < (B) < (C) < (D)

CA0089

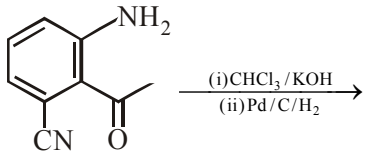
node06\B0BC BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbon\01_Th & Ex.p65

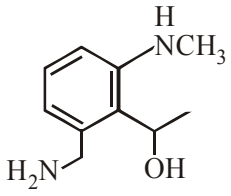
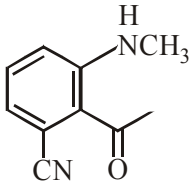
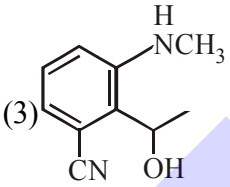
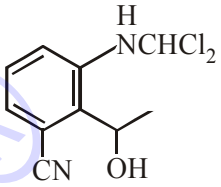
16. Which of the following amines can be prepared by Gabriel phthalimide reaction ?
[JEE Main (Apr)-2019]

(1) Neo-pentylamine (2) n-butylamine (3) triethylamine (4) t-butylamine

CA0091

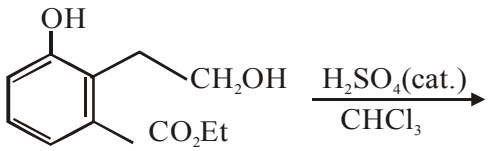
17. The major product obtained in the following reaction is : **[JEE Main (Apr)-2019]**

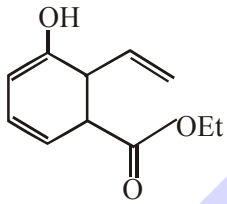
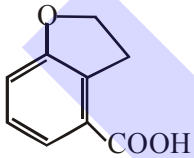
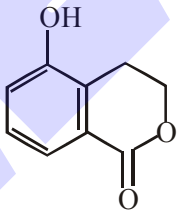
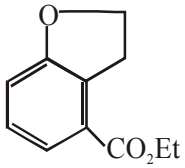


(1) 
(2) 
(3) 
(4) 

CA0092

18. The major product of the following reaction is: **[JEE Main (Apr)-2019]**



(1) 
(2) 
(3) 
(4) 

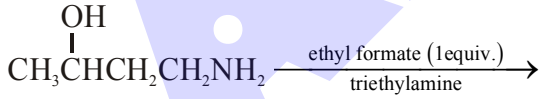
CA0093

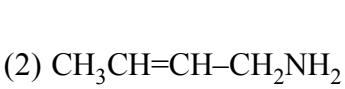
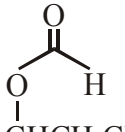
19. Hinsberg's reagent is : **[JEE Main (Apr)-2019]**

(1) $C_6H_5SO_2Cl$ (2) C_6H_5COCl (3) $SOCl_2$ (4) $(COCl)_2$

CA0094

20. The major product of the following reaction is : **[JEE Main (Apr)-2019]**



(1) 
(2) 
(3) 
(4) 

CA0095

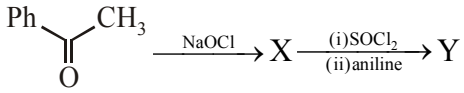
21. Ethylamine ($C_2H_5NH_2$) can be obtained from N-ethylphthalimide on treatment with :
[JEE Main (Apr)-2019]

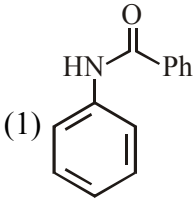
(1) $NaBH_4$ (2) CaH_2 (3) H_2O (4) NH_2NH_2

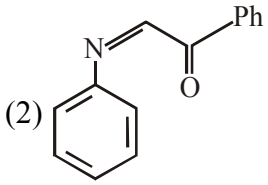
CA0096

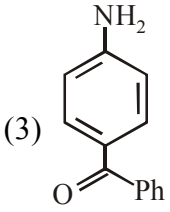
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

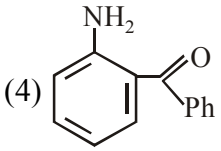
22. The major product 'Y' in the following reaction is:- **[JEE Main (Apr)-2019]**



(1) 

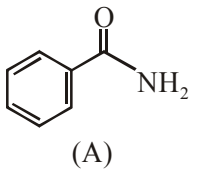
(2) 

(3) 

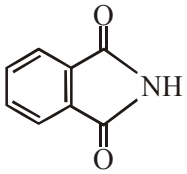
(4) 

CA0097

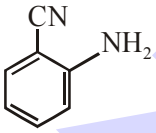
23. The increasing order of reactivity of the following compounds towards reaction with alkyl halides directly is : **[JEE-MAIN-(January) -2019]**



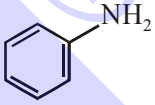
(A)



(B)



(C)



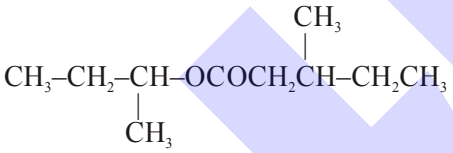
(D)

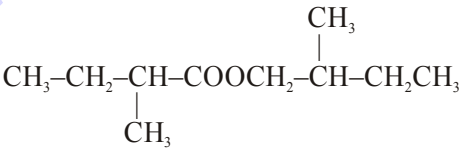
(1) (B) < (A) < (D) < (C) (2) (B) < (A) < (C) < (D)

(3) (A) < (C) < (D) < (B) (4) (A) < (B) < (C) < (D) **HD0180**

24. An organic compound [A], molecular formula $C_{10}H_{20}O_2$ was hydrolyzed with dilute sulphuric acid to give a carboxylic acid [B] and alcohol [C]. Oxidation of [C] with $CrO_3 - H_2SO_4$ produced [B]. Which of the following structures are not possible for [A] ? **[JEE-MAIN-2020]**

(1) $(CH_3)_3–C–COOCH_2C(CH_3)_3$ (2) $CH_3CH_2CH_2COOCH_2CH_2CH_2CH_3$

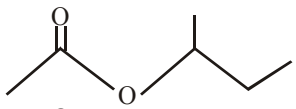
(3) 

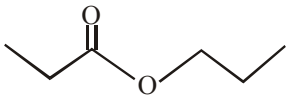
(4) 

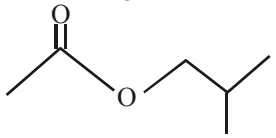
CA0121

25. An organic compound (A) (molecular formula $C_6H_{12}O_2$) was hydrolysed with dil. H_2SO_4 to give a carboxylic acid (B) and an alcohol (C). 'C' give white turbidity immediately when treated with anhydrous $ZnCl_2$ and conc. HCl. The organic compound (A) is : **[JEE-MAIN-2020]**

(1) 

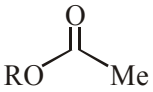
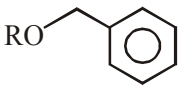
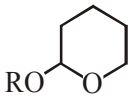
(2) 

(3) 

(4) 

CA0123

26. Which of the following derivatives of alcohols is unstable in an aqueous base ? **[JEE-MAIN-2020]**

(1) RO – CMe_3 (2)  (3)  (4) 

CA0124

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

27. Match list - I and List - II. **[JEE-MAIN-2021]**

List-I	**List-II**
(a) $R-\overset{\overset{O}{\|}}{C}-Cl \rightarrow R-CHO$	(i) $Br_2/NaOH$
(b) $R-CH_2-COOH \rightarrow R-\underset{\underset{Cl}{\|}}{CH}-COOH$	(ii) $H_2/Pd-BaSO_4$
(c) $R-\overset{\overset{O}{\|}}{C}-NH_2 \rightarrow R-NH_2$	(iii) Zn(Hg)/Conc.HCl
(d) $R-\overset{\overset{O}{\|}}{C}-CH_3 \rightarrow R-CH_2-CH_3$	(iv) Cl_2/Red P, H_2O

Choose the correct answer from the options given below :

(1) (a)–(ii), (b)–(i), (c)–(iv), (d)–(iii)
(2) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)
(3) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)
(4) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

CA0125

28. The major product of the following chemical reaction is : **[JEE-MAIN-2021]**

$$CH_3CH_2CN \xrightarrow[(3)\ Pd/BaSO_4,H_2]{(1)\ H_3O^+,\ \Delta \\ (2)\ SOCl_2} ?$$

(1) $CH_3CH_2CH_3$ (2) $CH_3CH_2CH_2OH$ (3) $(CH_3CH_2CO)_2O$ (4) CH_3CH_2CHO

CA0126

29. What is 'X' in the given reaction? **[JEE-MAIN-2021]**

$$\begin{matrix}CH_2OH\\|\\CH_2OH\end{matrix} + \text{oxalic acid} \xrightarrow{210°C} \underset{\text{(major product)}}{X}$$

(1) $\begin{matrix}CH_2\\\|\\CH_2\end{matrix}$ (2) $\begin{matrix}CH-OH\\\|\\CH_2\end{matrix}$ (3) $\begin{matrix}CHO\\|\\CHO\end{matrix}$ (4) $\begin{matrix}CH_2OH\\|\\CHO\end{matrix}$

CA0127

30. An amine on reaction with benzenesulphonyl chloride produces a compound insoluble in alkaline solution. This amine can be prepared by ammonolysis of ethyl chloride. The correct structure of amine is : **[JEE-MAIN-2021]**

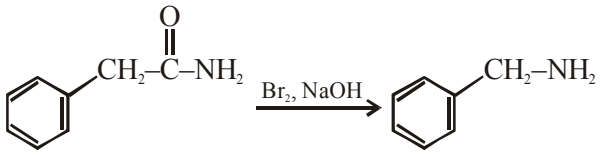
(1) 
(2) $CH_3CH_2NH_2$

(3) $CH_3CH_2CH_2NHCH_3$ (4) $CH_3CH_2CH_2\overset{\overset{H}{|}}{N}-CH_2CH_3$

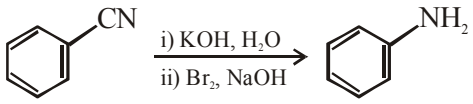
CA0128

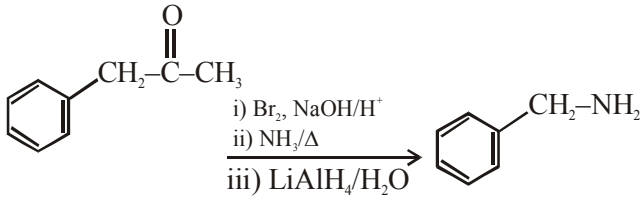
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

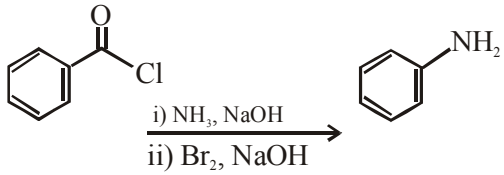
31. Which of the following reaction DOES NOT involve Hoffmann Bromamide degradation ?

(1) 

[JEE-MAIN-2021]

(2) 

(3) 

(4) 

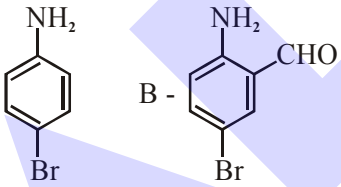
CA0129

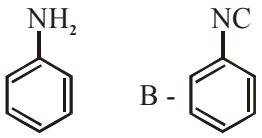
32. Ammonolysis of Alkyl halides followed by the treatment with NaOH solution can be used to prepare primary, secondary and tertiary amines. The purpose of NaOH in the reaction is : **[JEE-MAIN-2021]**

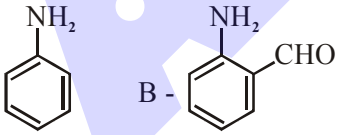
(1) to remove basic impurities
(2) to activate NH_3 used in the reaction
(3) to remove acidic impurities
(4) to increase the reactivity of alkyl halide

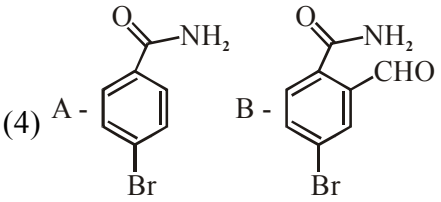
CA0130

33. Hoffmann bromomide degradation of benzamide gives product A, which upon heating with $CHCl_3$ and NaOH gives product B. The structures of A and B are : **[JEE-MAIN-2021]**

(1) 

(2) 

(3) 

(4) 

CA0131

34. Which of the following reaction is an example of ammonolysis? **[JEE-MAIN-2021]**

(1) $C_6H_5COCl + C_6H_5NH_2 \longrightarrow C_6H_5CONHC_6H_5$

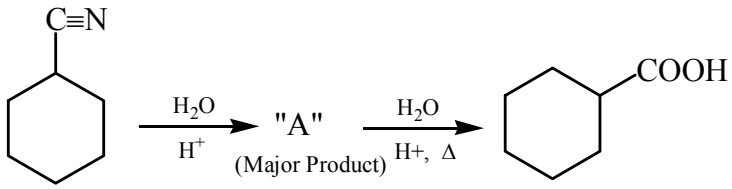
(2) $C_6H_5CH_2CN \xrightarrow{[H]} C_6H_5CH_2CH_2NH_2$

(3) $C_6H_5NH_2 \xrightarrow{HCl} C_6H_5\overset{+}{N}H_3Cl^-$

(4) $C_6H_5CH_2Cl + NH_3 \longrightarrow C_6H_5CH_2NH_2$

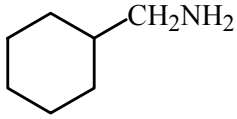
CA0132

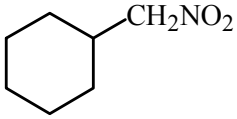
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biamole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

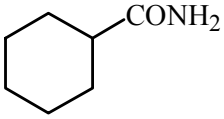
35. 

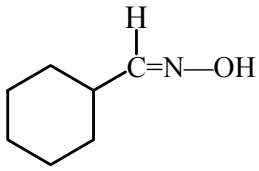
[JEE-MAIN-2021]

Consider the above chemical reaction and identify product "A"

(1) 

(2) 

(3) 

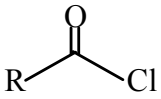
(4) 

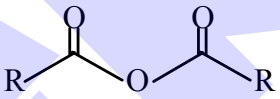
CA0133

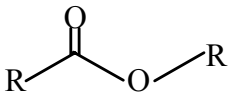
36. In the reaction of hypobromite with amide, the carbonyl carbon is lost as : **[JEE-MAIN-2021]**

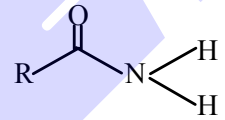
(1) CO_3^{2-} (2) HCO_3^- (3) CO_2 (4) CO

CA0134

37. (A) 

(B) 

(C) 

(D) 

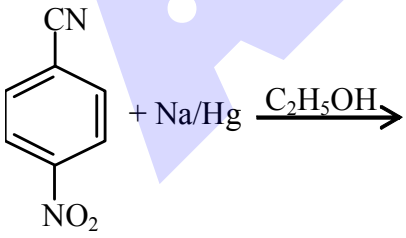
The correct order of their reactivity towards hydrolysis at room temperature is :

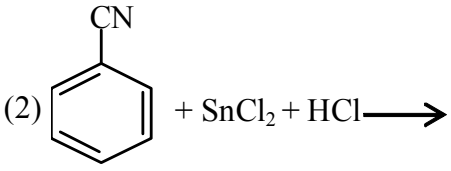
(1) (A) > (B) > (C) > (D) (2) (D) > (A) > (B) > (C) **[JEE-MAIN-2021]**

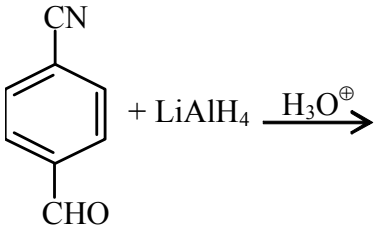
(3) (D) > (B) > (A) > (C) (4) (A) > (C) > (B) > (D)

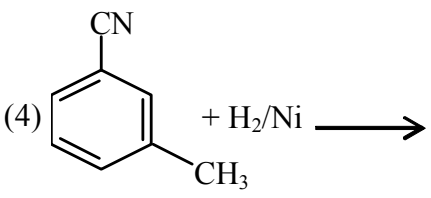
CA0135

38. Which one of the products of the following reactions does not react with Hinsberg reagent to form sulphonamide? **[JEE-MAIN-2021]**

(1) 

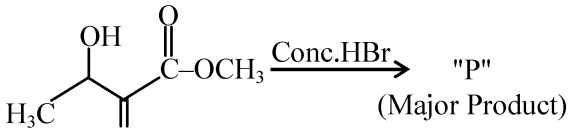
(2) 

(3) 

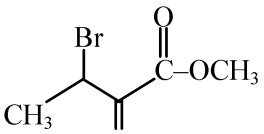
(4) 

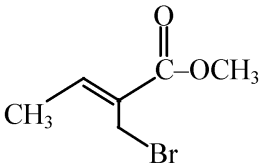
CA0137

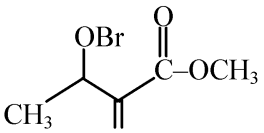
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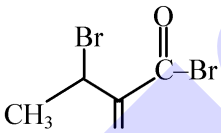
39. 

Consider the above reaction, the major product "P" formed is :- **[JEE-MAIN-2021]**

(1) 

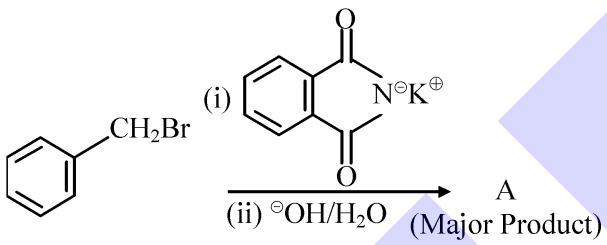
(2) 

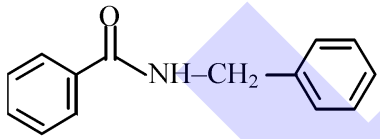
(3) 

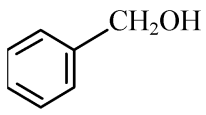
(4) 

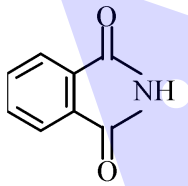
CA0138

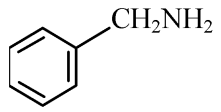
40. What is A in the following reaction ? **[JEE-MAIN-2021]**



(1) 

(2) 

(3) 

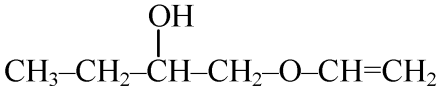
(4) 

CA0139

41. In the following sequence of reactions a compound A, (molecular formula $C_6H_{12}O_2$) with a straight chain structure gives a C_4 carboxylic acid. A is : **[JEE-MAIN-2021]**

$$\mathbf{A} \xrightarrow[H_3O^+]{LiAlH_4} \mathbf{B} \xrightarrow{\text{Oxidation}} C_4 - \text{carboxylic acid}$$

(1) $CH_3 - CH_2 - COO - CH_2 - CH_2 - CH_3$

(2) 

(3) $CH_3 - CH_2 - CH_2 - COO - CH_2 - CH$

(4) $CH_3 - CH_2 - CH_2 - O - CH = CH - CH_2 - OH$

CA0140

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42. Match List-I with List-II : **[JEE-MAIN-2021]**

List-I (Chemical Reaction)	**List-II (Reagent used)**
(a) $CH_3COOCH_2CH_3 \rightarrow CH_3CH_2OH$	(i) CH_3MgBr / H_3O^+ (1.equivalent)
(b) $CH_3COOCH_3 \rightarrow CH_3CHO$	(ii) H_2SO_4 / H_2O
(c) $CH_3C \equiv N \rightarrow CH_3CHO$	(iii) DIBAL-H/H_2O
(d) $CH_3C \equiv N \rightarrow CH_3\text{-}C(=O)\text{-}CH_3$	(iv) $SnCl_2$, HCl/H_2O

Choose the most appropriate match :

(1) a-ii, b-iv, c-iii, d-i

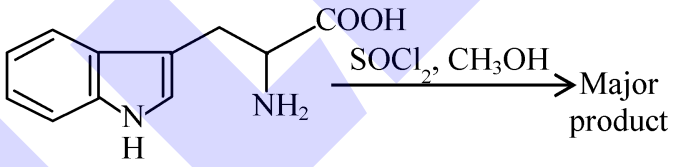
(2) a-iv, b-ii, c-iii, d-i

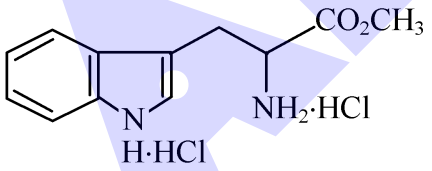
(3) a-ii, b-iii, c-iv, d-i

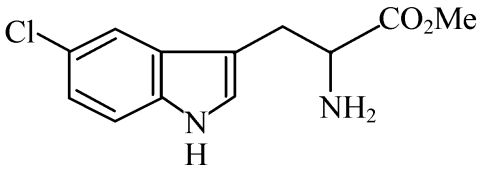
(4) a-iii, b-ii, c-i, d-iv

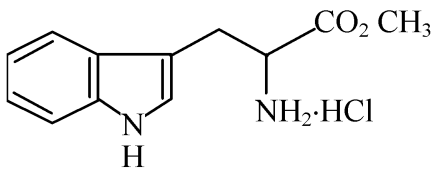
CA0141

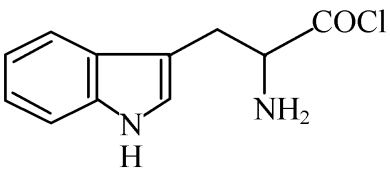
43. The major product formed in the following reaction is : **[JEE-MAIN-2021]**



(1) 

(2) 

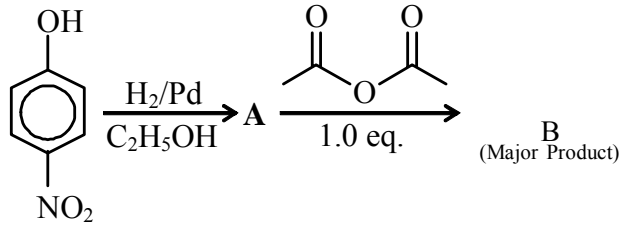
(3) 

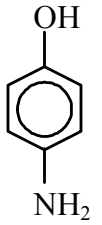
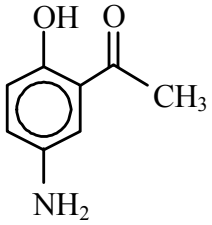
(4) 

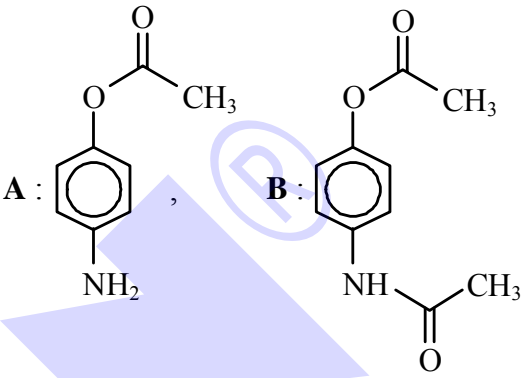
CA0143

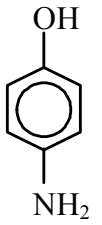
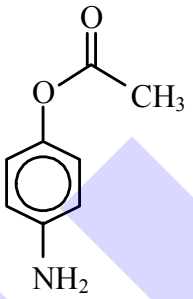
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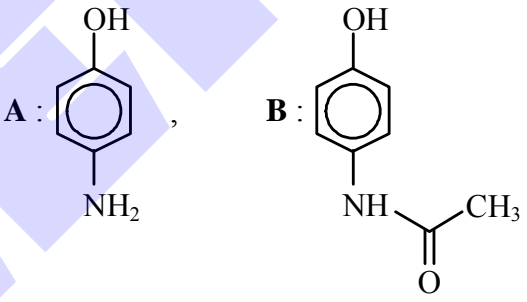
44. The correct structures of A and B formed in the following reactions are : **[JEE-MAIN-2021]**



(1) **A** :  , **B** : 

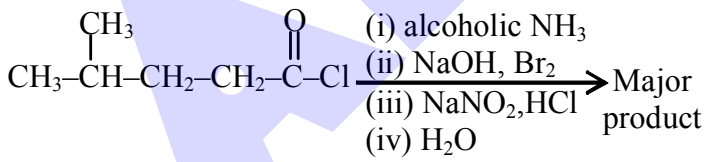
(2) 

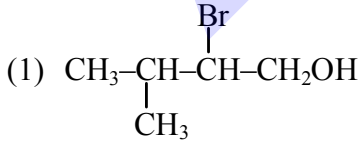
(3) **A** :  , **B** : 

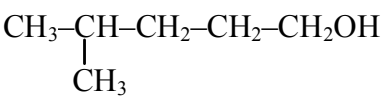
(4) 

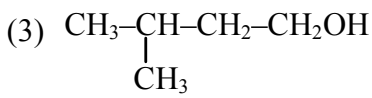
CA0144

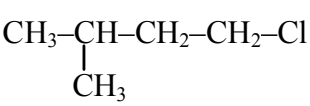
45. The major product of the following reaction is : **[JEE-MAIN-2021]**



(1) 

(2) 

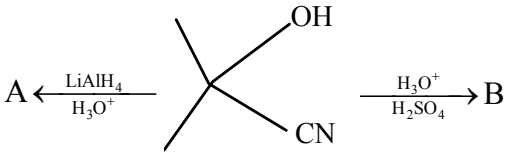
(3) 

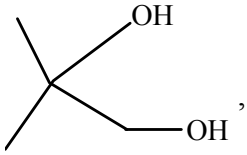
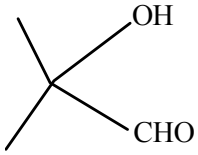
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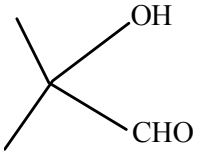
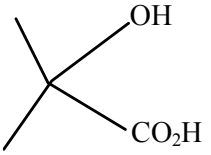
CA0145

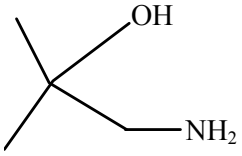
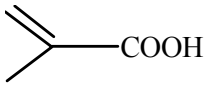
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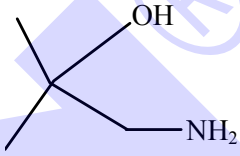
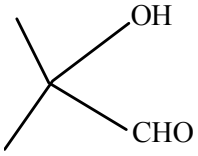
46. The major products A and B in the following set of reactions are : **[JEE-MAIN-2021]**



(1) A =  , B = 

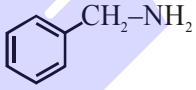
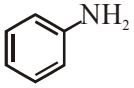
(2) A =  , B = 

(3) A =  , B = 

(4) A =  , B = 

CA0146

47. The total number of amines among the following which can be synthesized by Gabriel synthesis is _____. **[JEE-MAIN-2021]**

(i)  (ii) $CH_3CH_2NH_2$ (iii)  (iv) 

CA0148

48. The number of compound/s given below which contain/s –COOH group is _____. **[JEE-MAIN-2021]**

(i) Sulphanilic acid (ii) Picric acid (iii) Aspirin (iv) Ascorbic acid

CA0149

49. A reaction of 0.1 mole of Benzylamine with bromomethane gave 23 g of Benzyl trimethyl ammonium bromide. The number of moles of bromomethane consumed in this reaction are $n \times 10^{-1}$, when n = _____. (Round off to the Nearest Integer). **[JEE-MAIN-2021]**

(Given : Atomic masses : C : 12.0 u, H : 1.0 u, N : 14.0 u, Br : 80.0 u]

CA0150

50. Two statements are given below : **[JEE-MAIN-2022]**

Statement I: The melting point of monocarboxylic acid with even number of carbon atoms is higher than that of with odd number of carbon atoms acid immediately below and above it in the series.

Statement II : The solubility of monocarboxylic acids in water decreases with increase in molar mass.

Choose the most appropriate option:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct. **CA0180**

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51. Given below are two statements : **[JEE-MAIN-2022]**

Statement I : The esterification of carboxylic acid with an alcohol is a nucleophilic acyl substitution.

Statement II : Electron withdrawing groups in the carboxylic acid will increase the rate of esterification reaction.

Choose the most appropriate option :

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct. **CA0181**

52. The reaction of $R-\underset{\underset{O}{\|}}{C}-NH_2$ with bromine and KOH gives RNH_2 as the end product. Which one of the following is the intermediate product formed in this reaction ? **[JEE-MAIN-2022]**

(1) $R-\underset{\underset{O}{\|}}{C}-NH-Br$ (2) R-NH-Br (3) R-N=C=O (4) $R-\underset{\underset{O}{\|}}{C}-NBr_2$

CA0182

53. Given below are two statements:

Statements-I : In Hofmann degradation reaction, the migration of only an alkyl group takes place from carbonyl carbon of the amide to the nitrogen atom.

Statement-II : The group is migrated in Hofmann degradation reaction to electron deficient atom.

In the light of the above statement, choose the most appropriate answer from the options given below: **[JEE-MAIN-2022]**

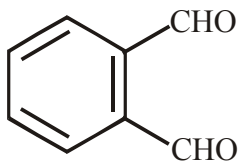
(1)Both Statement-I and Statement-II are correct

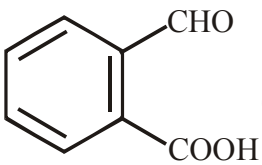
(2) Both Statement-I and Statement-II are incorrect

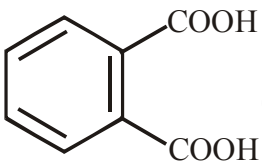
(3) Statement-I is correct but Statement-II is incorrect

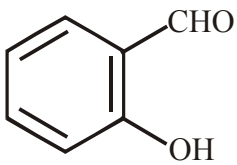
(4) Statement-I is incorrect but Statement-II is correct **CA0183**

54. An organic compound 'A' on reaction with NH_3 followed by heating gives compound B. Which on further strong heating gives compound C ($C_8H_5NO_2$). Compound C on sequential reaction with ethanolic KOH, alkyl chloride and hydrolysis with alkali gives a primary amine. The compound A is : **[JEE-MAIN-2022]**

(1) 

(2) 

(3) 

(4) 

CA0184

node06\B0BC BD\Kota\JEE(Advanced)\Enthuse\Chem Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbon\01_Th & Ex.p65

55. Match List I with List II **[JEE-MAIN-2022]**

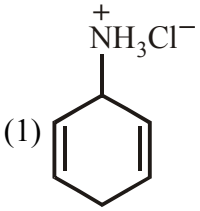
List I	List II
A. Benzenesulphonyl chloride	I. Test for primary amines
B. Hoffmann bromamide reaction	II. Anti Saytzeff
C. Carbylamine reaction	III. Hinsberg reagent
D. Hoffmann orientation	IV. Known reaction of Isocyanates.

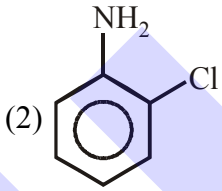
Choose the correct answer from the options given below:

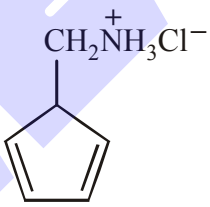
(1) A-IV, B -III, C-II, D-I (2) A-IV, B -II, C-I, D-III

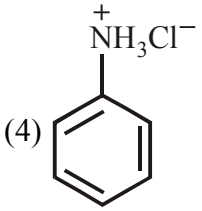
(3) A-III, B -IV, C-I, D-II (4) A-IV, B -III, C-I, D-II **CA0185**

56. An organic compound 'A' contains nitrogen and chlorine. It dissolves readily in water to give a solution that turns litmus red. Titration of compound 'A' with standard base indicates that the molecular weight of 'A' is 131 2. When a sample of 'A' is treated with aq. NaOH, a liquid separates which contains N but not Cl. Treatment of the obtained liquid with nitrous acid followed by phenol gives orange precipitate. The compound 'A' is : **[JEE-MAIN-2022]**

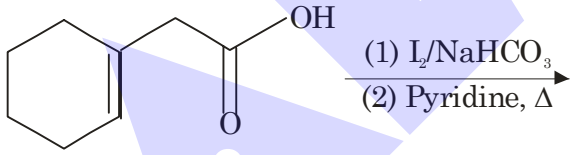
(1) 

(2) 

(3) 

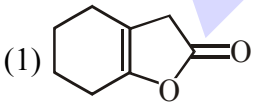
(4) 

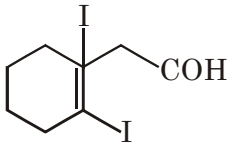
CA0186

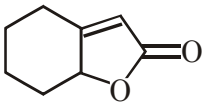
57. 

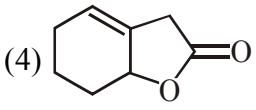
[JEE-MAIN-2022]

Find out the major product for the above reaction.

(1) 

(2) 

(3) 

(4) 

CA0187

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EXERCISE # JEE ADVANCED

1. Which of the following carboxylic acids undergo decarboxylation easily: **[IIT 1995]**

(A) $C_6H_5CO–CH_2COOH$ (B) $C_6H_5COCOOH$

(C) $C_6H_5\underset{\underset{OH}{|}}{CH_2} - COOH$ (D) $C_6H_5\underset{\underset{NH_2}{|}}{CH_2} - COOH$

CA0098

2. The molecular weight of benzoic acid in benzene as determined by depression in freezing point method corresponds to : **[IIT 1996]**

(A) Ionization of benzoic acid (B) Dimerisation of benzoic acid

(C) Trimerisation of benzoic acid (D) Solvation of benzoic acid

CA0099

3. When propionic acid is treated with aqueous $NaHCO_3$, CO_2 is liberated. The 'C' of CO_2 comes from.

(A) Methyl group (B) Carboxylic acid group

(C) methylene group (D) bicarbonate **[IIT 1999]**

CA0100

4. Benzoyl chloride is prepared from benzoic acid by: **[IIT 2000]**

(A) Cl_2, hν (B) SO_2Cl_2 (C) $SOCl_2$ (D) Cl_2, H_2O

CA0101

5. Which of the following acids has the smallest dissociation constant? **[IIT 2002]**

(A) $CH_3CHFCOOH$ (B) FCH_2CH_2COOH

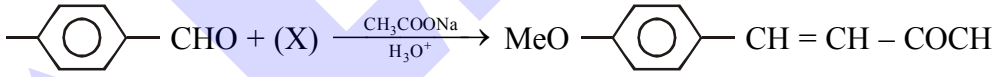
(C) $BrCH_2CH_2COOH$ (D) $CH_3CHBrCOOH$

CA0102

6. When benzamide is treated with $POCl_3$, the product is: **[IIT 2004]**

(A) Benzonitrile (B) Aniline (C) Chlorobenzene (D) Benzylamine

CA0103

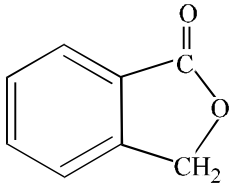
7. 

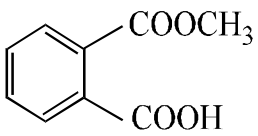
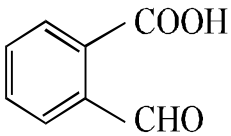
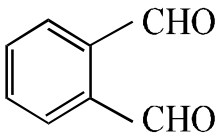
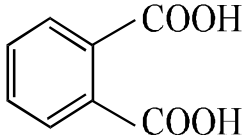
The compound (X) is **[IIT 2005]**

(A) $(CH_3CO)_2O$ (B) Br CH_2 – COOH (C) CH_3COOH (D) CHO – COOH

CA0104

8. Which of the following reactants on reaction with conc. NaOH followed by acidification gives the following lactone as the only product? **[IIT 2006]**



(A)  (B)  (C)  (D) 

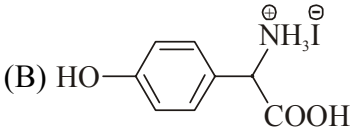
CA0105

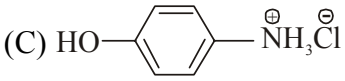
node06\B0BC BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

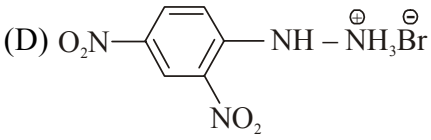
9. Match the compounds in **Column I** with their characteristic test(s)/reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS. **[IIT 2008]**

Column-I,

(A) $H_2N - \overset{\oplus}{N}H_3\overset{\ominus}{Cl}$

(B) 

(C) 

(D) 

Column-II.

(P) Sodium fusion extract of the compound gives Prussian blue colour with $FeSO_4$

(Q) Gives positive $FeCl_3$ test

(R) Gives white precipitate with $AgNO_3$

(S) Reacts with aldehydes to form the corresponding hydrazone derivative

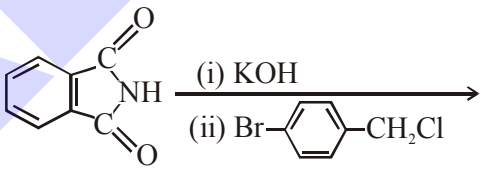
CA0106

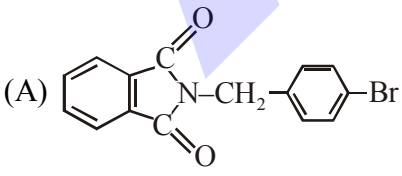
10. Match each of the compound in Column I with its characteristic reaction(s) in Column II. **[IIT 2009]**

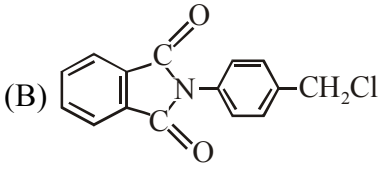
Column-I	**Column-II**
(A) $CH_3CH_2CH_2CN$	(P) Reduction with Pd–C / H_2
(B) $CH_3CH_2OCOCH_3$	(Q) Reduction with $SnCl_2$ / HCl
(C) $CH_3 - CH = CH - CH_2OH$	(R) Development of foul smell on treatment with chloroform and alcoholic KOH
(D) $CH_3CH_2CH_2CH_2NH_2$	(S) Reduction with diisobutylaluminium hydride (DIBAL-H)
	(T) Alkaline hydrolysis

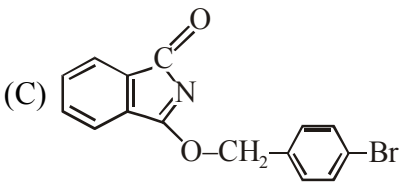
CA0107

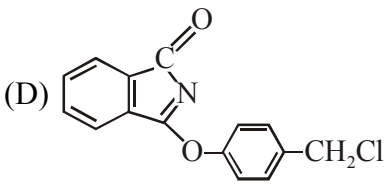
11. The major product of the following reaction is **[IIT 2011]**



(A) 

(B) 

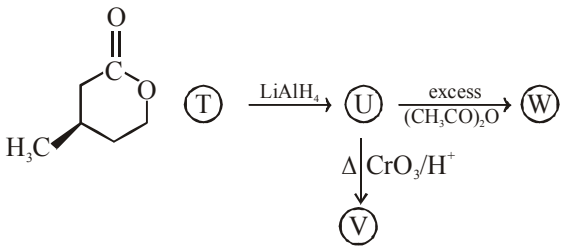
(C) 

(D) 

CA0108

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep, Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

12. With reference the scheme given, which of the given statement(s) about T, U, V & W is/are correct **[IIT 2012]**



(A) 'T' is soluble in hot aq NaOH

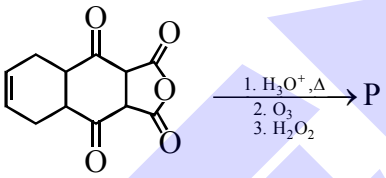
(B) 'U' is optically active

(C) mol formula of W is $C_{10}H_{18}O_4$

(D) V gives effervescence with aq $NaHCO_3$

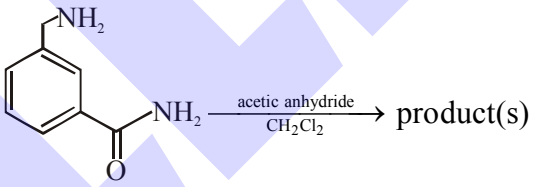
CA0109

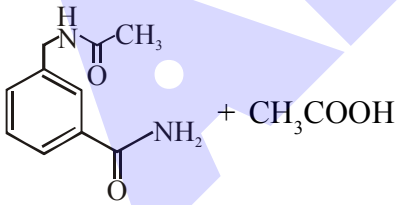
13. The total number of carboxylic acid groups in the product P is **[IIT 2013]**

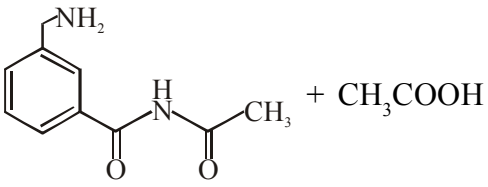


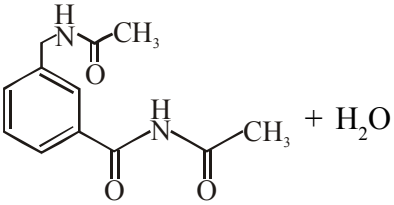
CA0111

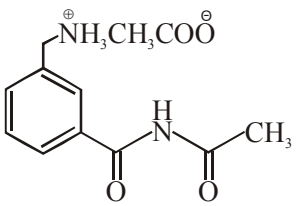
14. In the reaction shown below, the major product(s) formed is / are : **[IIT 2014]**



(A) 

(B) 

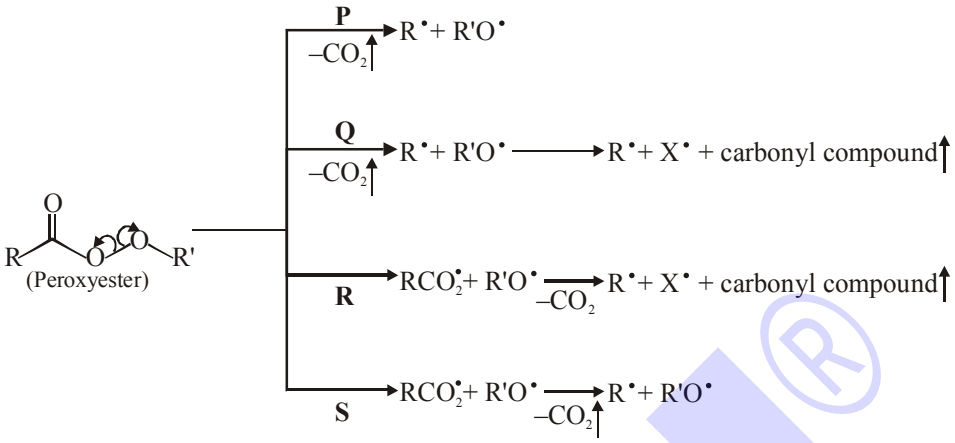
(C) 

(D) 

CA0112

node06\B0BC BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

15. Different possible **thermal** decomposition pathways for peroxyesters are shown below. Match each pathway from List-I with an appropriate structure from List-II and select the correct answer using the code given below the lists. **[IIT 2014]**



List-I

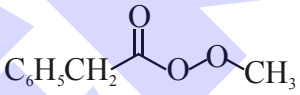
(P) Pathway **P**

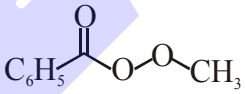
(Q) Pathway **Q**

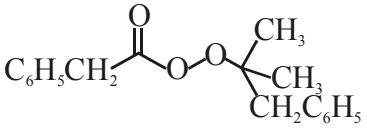
(R) Pathway **R**

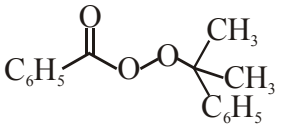
(S) Pathway **S**

List-II

(1) 

(2) 

(3) 

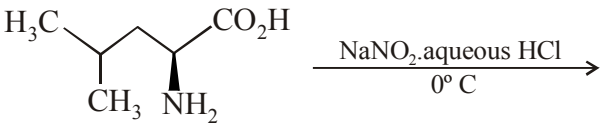
(4) 

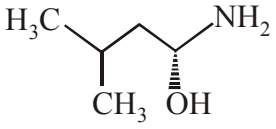
Code :

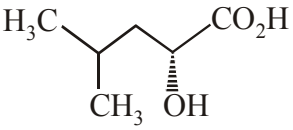
	P	Q	R	S
(A)	1	3	4	2
(B)	2	4	3	1
(C)	4	1	2	3
(D)	3	2	1	4

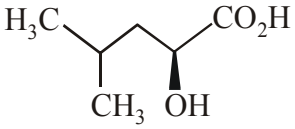
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

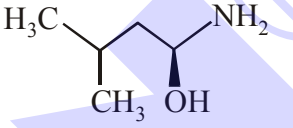
16. The major product of the reaction is : **[IIT 2015]**



(A) 

(B) 

(C) 

(D) 

CA0114

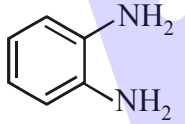
PARAGRAPH FOR NO. 17 & 18

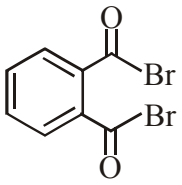
Treatment of compound **O** with $KMnO_4$ / H^+ gave **P**, which on heating with ammonia gave The compound **Q** on treatment with Br_2 / NaOH produced **R**. On strong heating, **Q** gave **S**, which on further treatmenet with ethyl 2-bromopropanoate in the presence of KOH following by acidification, gave a compound **T**. **[IIT-JEE-2016]**

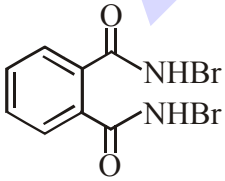


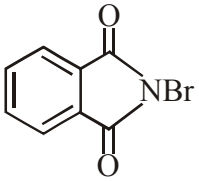
(O)

17. The compound **R** is :

(A) 

(B) 

(C) 

(D) 

CA0115

18. The compound **T** is :

(A) Glycine (B) Alanine (C) Valine (D) Serine

CA0116

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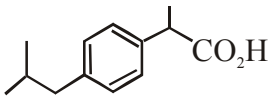
PARAGRAPH FOR NO. 19 & 20

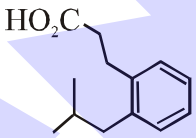
An organic acid P ($C_{11}H_{12}O_2$) can easily be oxidized to a dibasic acid which reacts with ethyleneglycol to produce a polymer dacron. Upon ozonolysis, **P** gives an aliphatic ketone as one of the products. **P** undergoes the following reaction sequences to furnish **R** via The compound **P** also undergoes another set of reactions to produce **S.** **[IIT-JEE(Adv.)-2018]**

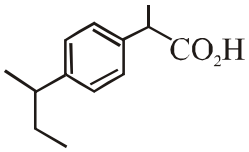
$$S \xleftarrow[(4)\ CHCl_3,\ KOH,\ \Delta]{\substack{(1)\ H_2/Pd\text{-}C\\(2)\ NH_3/\Delta\\(3)\ Br_2/NaOH\\(5)\ H_2/Pd\text{-}C}} P \xrightarrow[(3)\ MeMgBr,\ CdCl_2]{\substack{(1)\ H_2/Pd\text{-}C\\(2)\ SOCl_2\\(4)\ NaBH_4}} Q \xrightarrow[(3)\ CO_2\ (dry\ ice)]{\substack{(1)\ HCl\\(2)\ Mg/Et_2O\\(4)\ H_3O^+}} R$$

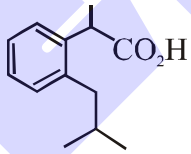
(There are two questions based on PARAGRAPH "A", the question given below is one of them)

19. The compound **R** is

(A) 

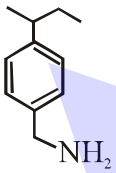
(B) 

(C) 

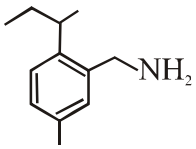
(D) 

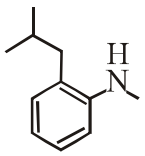
CA0118

20. The compound **S** is

(A) 

(B) 

(C) 

(D) 

CA0119

Question Stem for Question Nos. 21 and 22

Reaction of x g of Sn with HCl quantitatively produced a salt. Entire amount of the salt reacted with y g of nitrobenzene in the presence of required amount of HCl to produce 1.29 g of an organic salt (quantitatively).

(Use Molar masses (in g mol-1) of H, C, N, O, Cl and Sn as 1, 12, 14, 16, 35 and 119, respectively).

21. The value of x is _____. **[IIT-JEE(Adv.)-2021]**

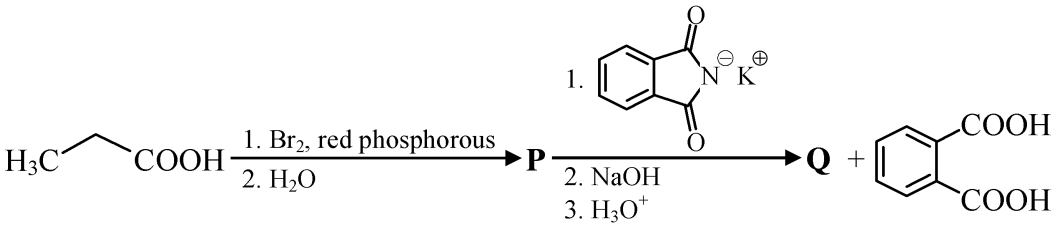
CA0188

22. The value of y is _____. **[IIT-JEE(Adv.)-2021]**

CA0189

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23. Considering the reaction sequence given below, the correct statement(s) is(are)



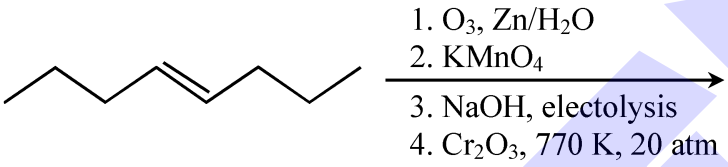
(A) P can be reduced to a primary alcohol using $NaBH_4$. **[IIT-JEE(Adv.)-2022]**

(B) Treating P with conc. NH_4OH solution followed by acidification gives Q.

(C) Treating Q with a solution of $NaNO_2$ in aq. HCl liberates N_2.

(D) P is more acidic than CH_3CH_2COOH. **CA0190**

24. The number of $-CH_2-$ (methylene) groups in the product formed from the following reaction sequence is _____. **[IIT-JEE(Adv.)-2022]**



CA0191

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ANSWER KEY

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	D	C	C	B	B	C	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	B	B	A	C	A	A	C
Que.	21	22	23	24	25	26	27	28	29	
Ans.	B	C	D	C	D	C	C	D	B	

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	B	A	C	D	D	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	B	C	B	D	B	A	D	A
Que.	21	22	23	24	25					
Ans.	D	C	C	B	D					

EXERCISE # S-I

Que.	1	2	3	4	5	6	7	8	
Ans.	A,C	A,B	A	A,C	A,B,C	C,D	A→P,S; B→Q,S; C→P,S; D→R;	A→S; B→R; C→P; D→Q;	
Que.	9		10		11		12	13	14
Ans.	A→S; B→R; C→P; D→Q;		A→R; B→S; C→Q; D→P;		A→P; B→Q; C→S; D→R,S;		A→P; B→R; C→P,S; D→Q;	4	5
Que.	15	16	17	18	19				
Ans.	9	Theoritical	Theoritical	Theoritical	18				

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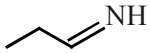
EXERCISE # JEE-MAIN

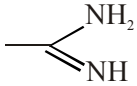
1. (4) **2. (4)** **3. (1)** **4. (2)**

5. (1) **6. (1)**

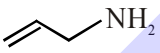
7. (2)

Sol. Order of base nature depends on electron donation tendency.

In compound  nitrogen is sp^2 hybridized so least basic among all given compound.

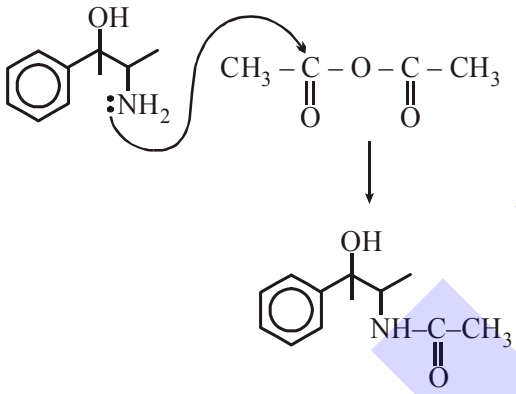
compound  is very strong nitrogeneous organic base as lone pair of one nitrogen delocalize in resonance and make another nitrogen negativly charged and conjugate acid have two equivalent resonating structure.

Thus it is most basic in given compouds.

 (secondary amine) more basic than  (primary amine)

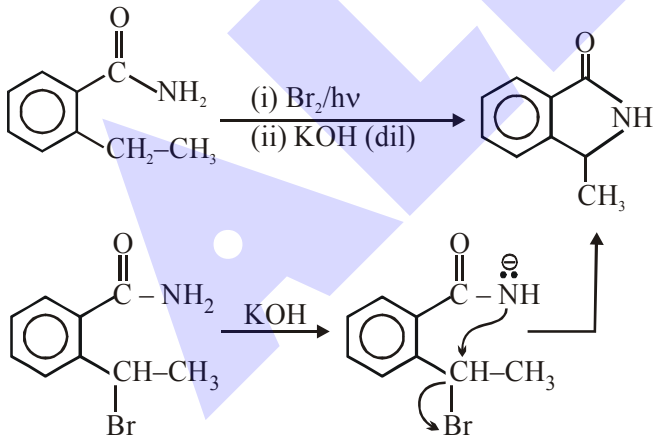
8. (3)

Sol.



9. (3)

Sol.



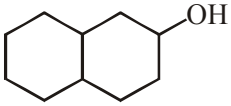
10. (4)

Sol. Adipic acid $CO_2H–(CH_2)_4–CO_2H \xrightarrow[\text{agent}]{\text{dehydrating}}$ 7 membered cyclic anhydride (Very unstable)

11. (2)

Sol. More is the electrophilic character of carbonyl group of ester faster is the alkaline hydrolysis.

12. (Bonus)

Sol. Answer should be 

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13. (3)

14. (3)

Sol.

$$[X] \xrightarrow[NaOH]{Br_2} C_3H_9N \xrightarrow[KOH]{CHCl_3} CH_3CH_2CH_2{-}NC$$

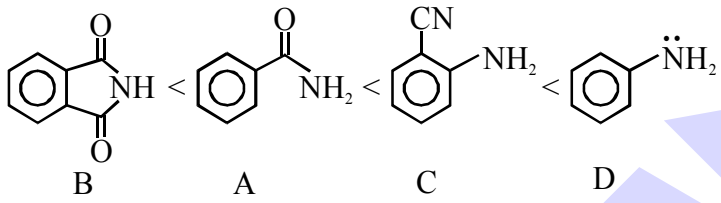
Hoff mann's Bromaide degradation — Carbylamine Reaction

Thus [X] must be aride with oen carbon more than is amine.

Thus [X] is $CH_3CH_2CH_2CONH_2$

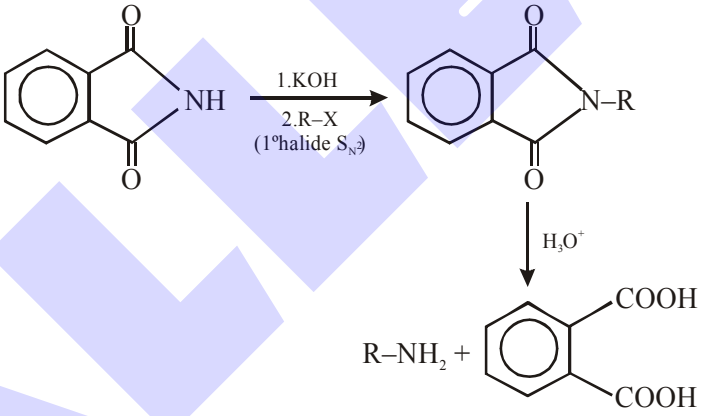
15. (2)

Sol. Nucleophilicity order



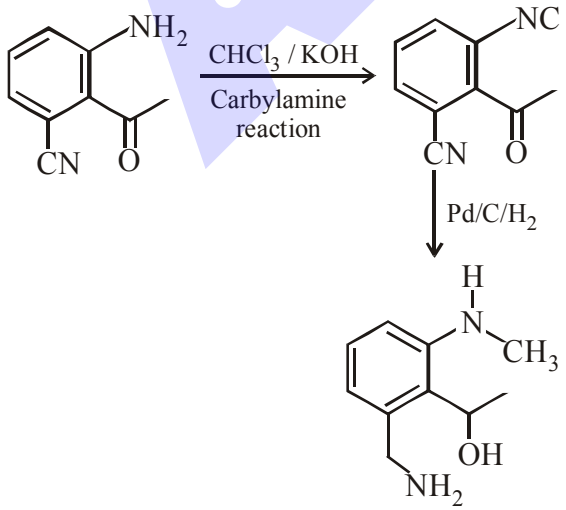
16. (2)

Sol. Gabriel phthalimide synthesis :



17. (1)

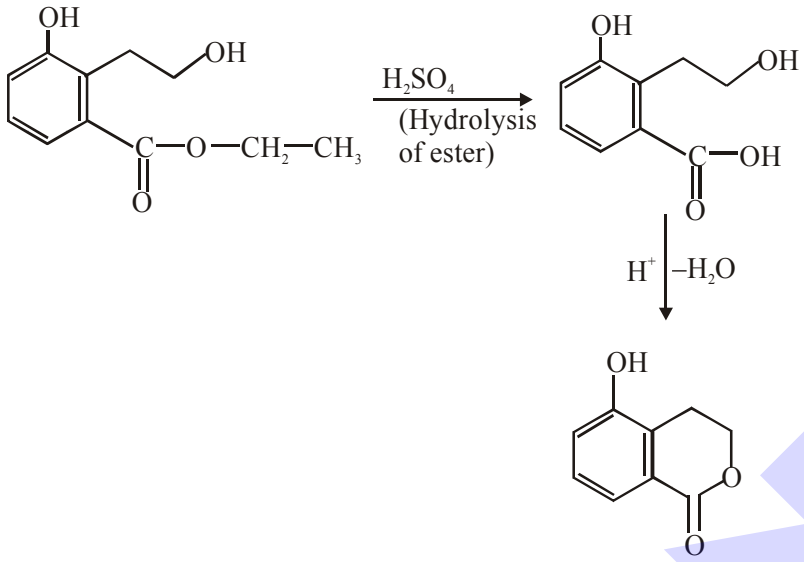
Sol.



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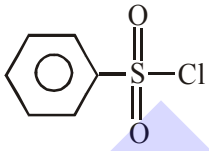
18. (3)

Sol.



19. (1)

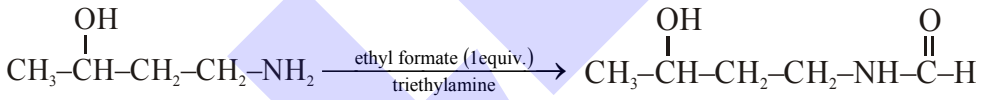
Sol. Hinsberg Reagenl ⇒



[Benzene Sulphonyl chloride]

20. (1)

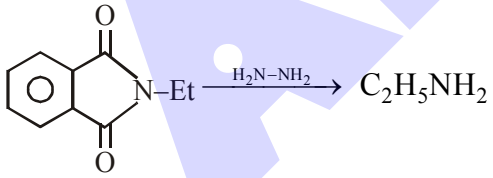
Sol.



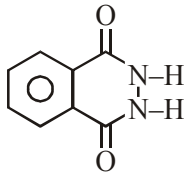
as NH_2 is a better nucleophile than OH.

21. (4)

Sol.



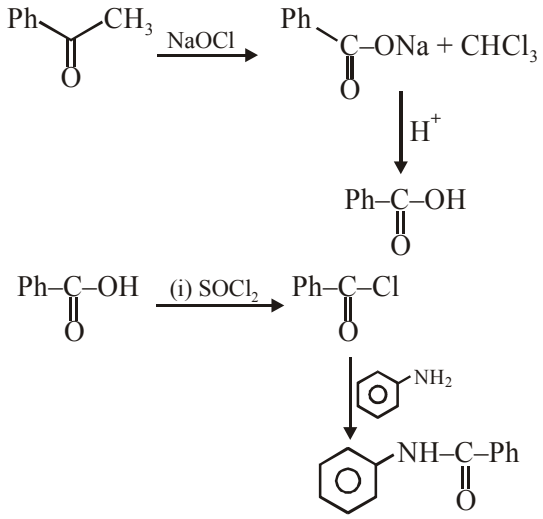
reagent is $NH_2–NH_2$ byproduct will be



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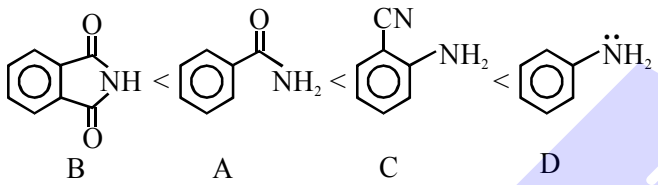
22. (1)

Sol.



23. (2) from halogen

Sol. Nucleophilicity order



B A C D

24. (3)	**25. (1)**	**26. (2)**	**27. (3)**
28. (4)	**29. (1)**	**30. (4)**	**31. (3)**
32. (3)	**33. (2)**	**34. (4)**	**35. (3)**
36. (1)	**37. (1)**	**38. (2)**	**39. (2)**
40. (4)	**41. (3)**	**42. (3)**	**43. (3)**
44. (4)	**45. (3)**	**46. (3)**	**47. (3)**
48. (1)	**49. (3)**		

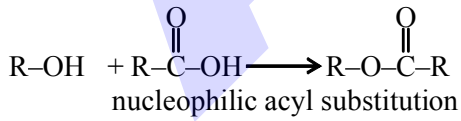
50. (1)

Sol. I . Better packing efficiency of monocarboxylic acids with even number of carbon atoms results in higher M.P

II. As molar mass increases hydrophobic part size increase hence solubility decreases.

51. (1)

Sol.



nucleophilic acyl substitution

electron with drawing group on carboxylic acid will increase the rate of esterification

52. Allen (1 & 3)

53. (4)

Sol. $R - CO - NH_2 + Br_2 + NaOH \rightarrow R - NH_2 + Na_2CO_3 + NaBr + H_2O$

$R - CO - NH_2 + \bar{O}H \rightarrow R - CO - \bar{N}H \xrightarrow{Br_2}$

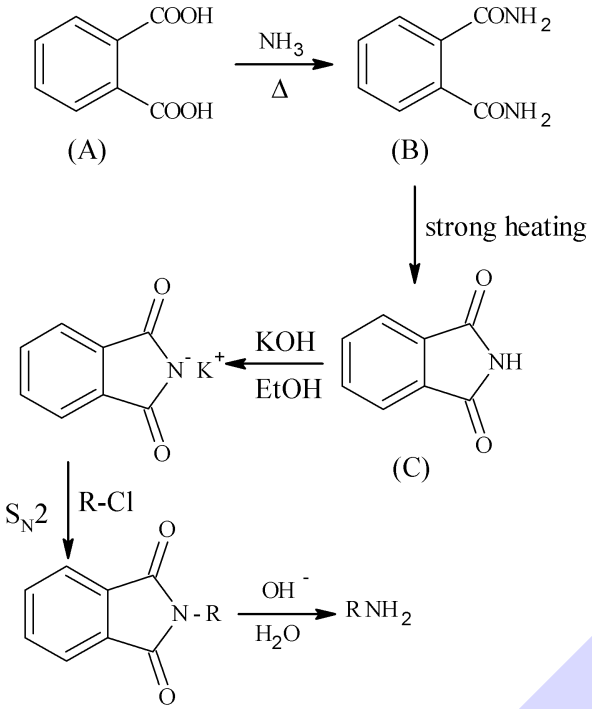
$R - CO - NH - Br \xrightarrow{OH^-} R - CO - \bar{N} - Br \xrightarrow{\text{migration of } \bar{R}} R - NCO \xrightarrow{2\bar{O}H} RNH_2 + CO_3^{2-}$

In this reaction of alkyl as well as aryl group can migrate to electron deficient nitrogen atom.

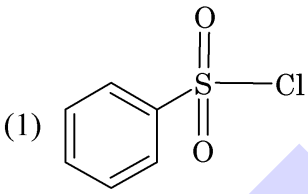
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

54. (3)

Sol. Gabriel Pthalimide reaction



55. (3)



→ Hinsberg reagent

Benzen sulphonyl chloride

(2) Hoffmann bromamide reaction → known reaction of isocynates

$R–CO–NH_2 + X_2 + 4NaOH \rightarrow R–NH_2 + 2NaX + Na_2CO_3 + 2H_2O$

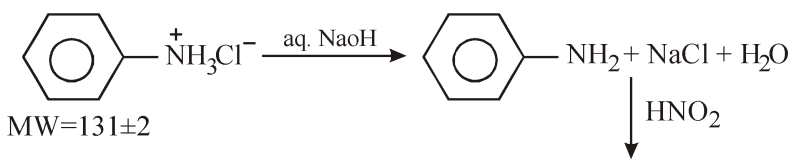
Intermediate : R- N = C = O (isocyanate)

(3) Carbylamine reaction → Test for primary amine

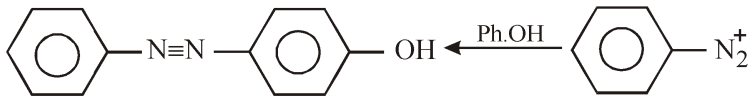
$R–NH_2$ or Ar - $NH_2 + CHCl_3 + 3KOH \rightarrow$ RNC or Ar - NC+ $3KCl + 3H_2O$

(4) Hoffmann orientation → Anti saytzeff (Formation of less substituted alkene as major product)

56. (4)



Sol.

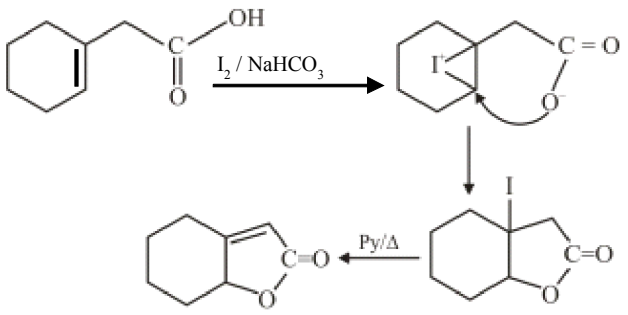


p-hydroxy azobenzene
(Orange dye)

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\5-Carboxylic-Aromatic-Biomole-Poly, POC & Sep. Tec-Che Every day\Eng\1-Carbox\01_Th & Ex.p65

57. (3)

Sol.



EXERCISE # JEE ADVANCED

1. (A) **2. (B)** **3. (D)** **4. (C)**

5. (C) **6. (A)** **7. (A)** **8. (C)**

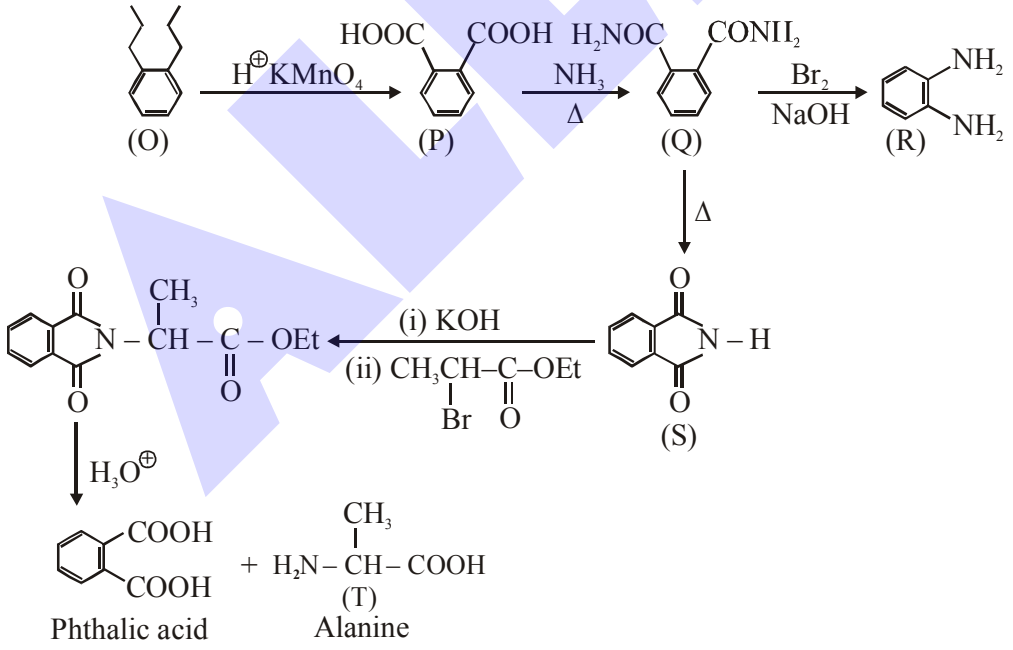
9. (A)→R, S ; (B)→P, Q ; (C)→P, Q, R ; (D)→P,S

10. (A)→P, Q, S, T ; (B)→P, S, T ; (C)→P ; (D)→R **11. (A)**

12. (A,C,D) **13. (2)** **14. (A)** **15. (A)**

16. (C) **17. (A)** **18. (B)**

Solution 17 & 18.



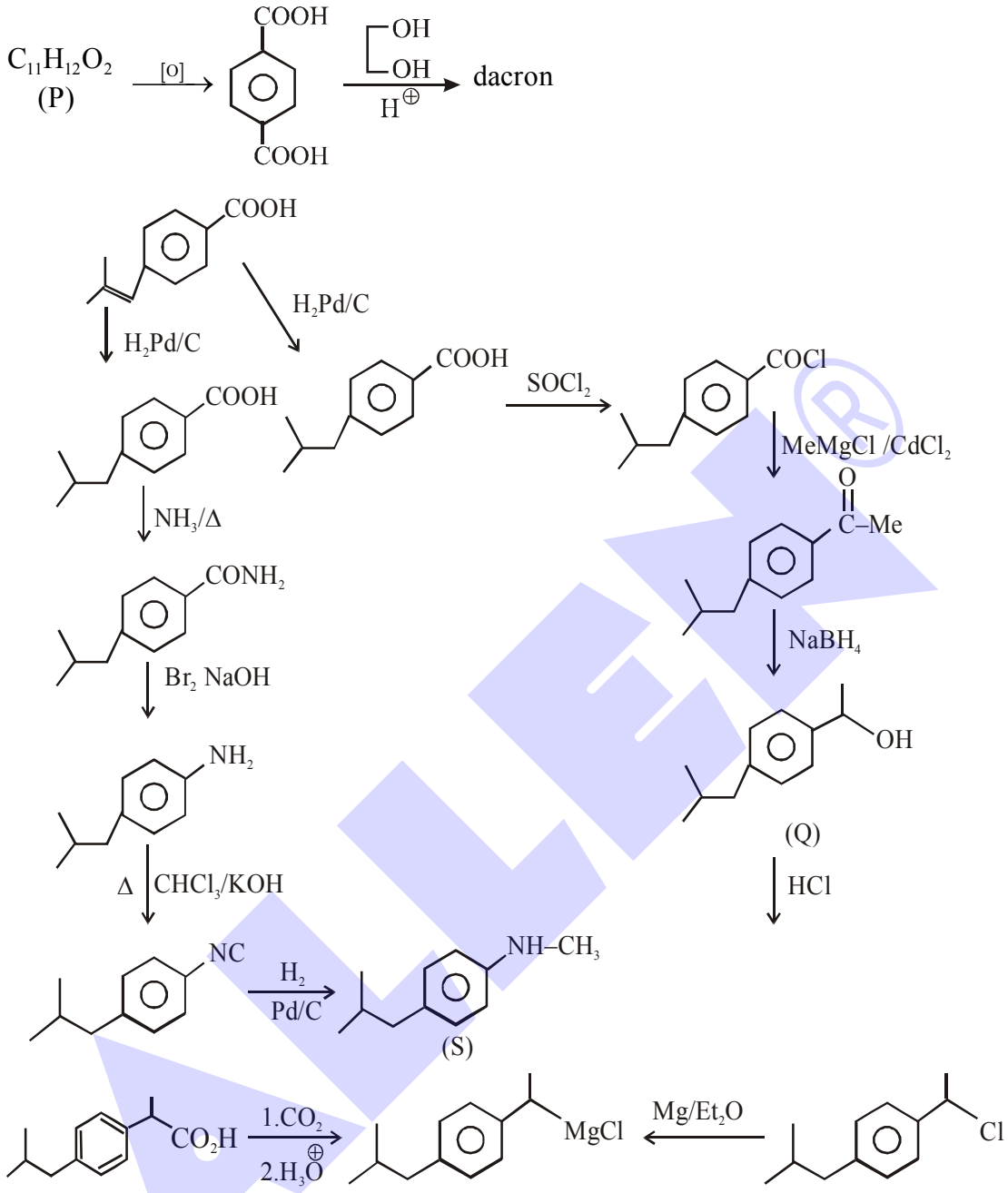
Q to R is Hoffmann's bromamide degradation reaction

S to T is Gabriel's phthalimide sysnthesis

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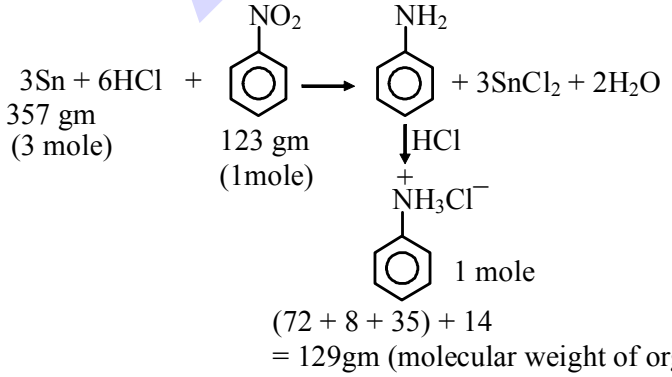
19. (A) **20. (B)**

Solution 19 & 20.



21. (3.57) **22. (1.23)**

Sol. The value of x and y are (Q.no. 21 and 22 combine solution)



(72 + 8 + 35) + 14

= 129gm (molecular weight of organic salt)

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So to get 1.29 gm organic salt.

We have to form 0.01 mole salt.

So 0.01 mole nitrobenzene is required.

0.03 mole Sn is required.

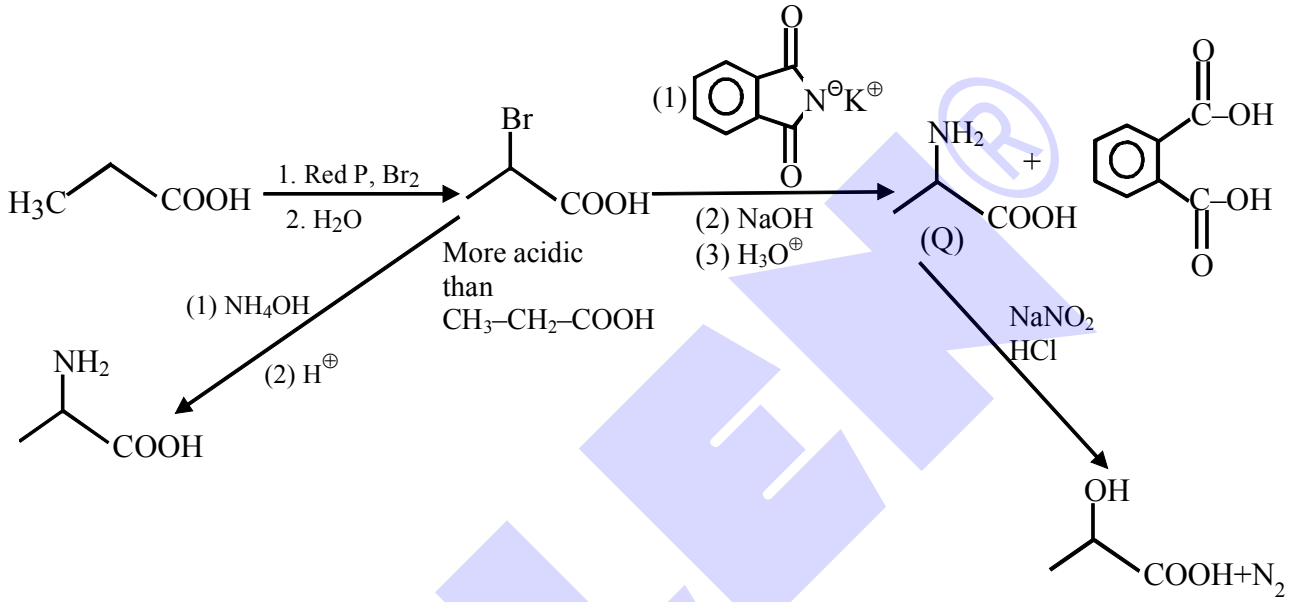
So the amount of nitrobenzene = $0.01 \times 123 = 1.23$ gm

the amount of Sn required = $0.01 \times 357 = 3.57$ gm

3.57 & 1.23

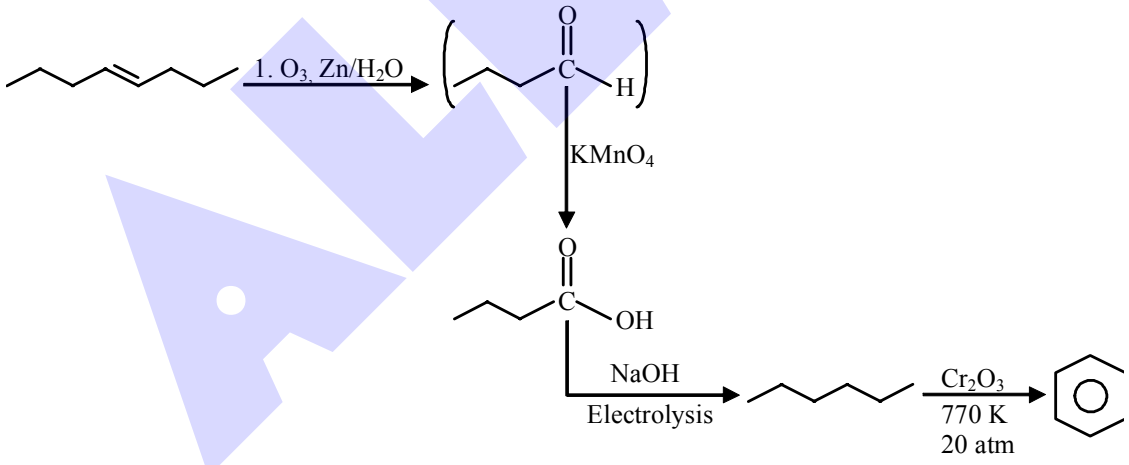
23. (B,C,D)

Sol.



24. (0)

Sol.



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